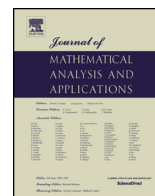




Contents lists available at ScienceDirect

Journal of Mathematical Analysis and Applications

journal homepage: www.elsevier.com/locate/jmaa

Regular Articles

Dynamical analysis of an age-structured dengue model with asymptomatic infection ☆

Xiaoguang Li^a, Liming Cai^{a,*}, Mohammad Murshed^b, Jin Wang^b^a College of Mathematics and Statistics, Xinyang Normal University, Xinyang 464000, PR China^b Department of Mathematics, University of Tennessee at Chattanooga, Chattanooga, TN 37403, USA

ARTICLE INFO

Article history:

Received 23 August 2022

Available online 20 February 2023

Submitted by A. Pugliese

Keywords:

Dengue

Age structure

Reproduction number

Stability

Backward bifurcation

ABSTRACT

In this paper, we propose a new mathematical model to investigate the transmission dynamics of dengue. The model is based on an age-structured system of differential and integral equations that couples the host and mosquito populations and that incorporates both symptomatic and asymptomatic infections. We derive the basic reproduction number and conduct a rigorous analysis on the local and global stabilities of the disease-free steady state. Meanwhile, we study the existence of the endemic steady states and explore conditions that could lead to the occurrence of a backward bifurcation. In addition, we establish the weak and strong uniform persistence properties of the system.

© 2023 Elsevier Inc. All rights reserved.

1. Introduction

Dengue, a mosquito-borne viral disease, has been a significant public health burden in the tropical and subtropical areas since the early 20th century [15]. The causative agent for this disease is dengue virus, a single positive-stranded RNA virus of the family *Flaviviridae*, with strains characterized by five serotypes [36]. Dengue virus is primarily transmitted by female mosquitoes of the *Aedes* genus that includes species such as *Aedes aegypti* and *Aedes albopictus* [31]. At present, it is estimated that around 400 million people are infected with dengue annually, and about half a million require hospital care [3,22,36].

Common symptoms of dengue include sudden-onset fever, headache, vomiting, and a characteristic skin rash. Severe cases of dengue infection can cause health complications such as dengue haemorrhagic fever and dengue shock syndrome, which may lead to death if not treated in a timely manner [12]. Individuals infected with dengue virus will go through an exposed (or, latent) period of 3–8 days without exhibiting any symptoms [16]. They become infectious after the latent period, but the majority (over 80%) of the

☆ Research is supported by the National Natural Science Foundation of China (No. 11871415, 12271466) and the Henan Province Distinguished Professor program.

* Corresponding author.

E-mail address: limingcai@amss.ac.cn (L. Cai).

cases with a primary infection remain asymptomatic [36]. These asymptomatic individuals can infect local mosquitoes which go on to bite other people and spread the infection to larger human populations. They also pose a threat to transfusion recipients through infected blood products [18]. Furthermore, asymptomatic individuals can be re-infected, referred to as a secondary infection, through mosquitoes carrying a different strain of dengue virus and become symptomatic with more severe health conditions [24]. The mechanism for the different viral serotypes to cause severe dengue disease is not entirely clear at present, but it is mostly linked to the Antibody-Dependent Enhancement (ADE), a phenomenon where pre-existing antibodies do not neutralize but rather enhance the new infection [2]. Hence, asymptomatic infectious individuals play a significant role in shaping the overall pattern of dengue transmission and spread. Quantifying such a role is essential in understanding the transmission dynamics of dengue and in designing effective prevention and intervention strategies.

Like many other infectious disease, the transmission of dengue exhibits strong heterogeneity among different human hosts. One of the most important factors to reflect such heterogeneity is the chronological age. Individuals of different ages may have different social behaviors, mobility levels, and contact dynamics, which subsequently lead to different infection risk, disease severity, survival capacity, and recovery ability. In particular, children are more vulnerable than adults because of their immature immune systems, and they have higher risk of severe complications for dengue [28,33]. Identifying such transmission characteristics associated with different age groups and classifying such an age structure in a dengue model are critical toward a holistic understanding of dengue transmission dynamics.

Numerous mathematical models have been published for the transmission dynamics of dengue [1,11–13,16,18–20,22,24,26,29,31]. For example, Esteva and Vargas studied the coexistence of different serotypes of dengue virus [11]. Pongsumpun and Samana published a model for the primary infection dynamics of dengue that involves both symptomatic and asymptomatic individuals [26]. Jan and Xiao modeled the effect of partial immunity from the susceptible human population on the transmission of dengue and performed an optimal control study for disease prevention and intervention [19]. Rashkov and Kooi integrated the vector dynamics of mosquitoes into the human transmission of dengue [29]. Rocha et al. considered the role of seasonality in dengue infection dynamics [30]. Ferguson et al. proposed a multi-strain model to investigate the effect of ADE on dengue transmission [13]. Aguiar et al. studied the differences between the primary and secondary infections of dengue in the presence of seasonality and multiple viral strains [1]. Kabir and Tanimoto conducted a cost-efficiency analysis for the effect of ADE on dengue vaccination [20]. In addition, a number of other mathematical models for dengue are surveyed in a recent review article [2]. There is no doubt that these modeling studies have substantially advanced our knowledge in the transmission dynamics of dengue and in the design of effective control strategies for this disease. However, the fundamental role of the asymptomatic infection in dengue transmission within a heterogeneous host population is only partially addressed at present. Particularly, our understanding of the intertwined relationship among the primary and secondary infections, the asymptomatic and symptomatic states, and the different age groups, remains limited.

In the present paper, we propose a new mathematical model for the transmission dynamics of dengue based on the age-structured modeling technique. Age-structured models have been employed in many epidemiological studies and have provided useful guidelines for disease control [5,8–10,12,21,22,25,27,32,35,37,38]. A few of these have been specifically devoted to dengue [12,22,27]. Our proposed study differs from the prior age-structured models in the integration of several factors that are important toward a deeper understanding of dengue dynamics. Our modeling framework will incorporate both mosquito-to-human and human-to-mosquito transmission routes, will include both symptomatic and asymptomatic infectious hosts, will place a special emphasis on the secondary infection from asymptomatic individuals, and will connect all these components in an age-structured setting. The model consists of a strongly nonlinear system of coupled differential and integral equations, which will be rigorously analyzed throughout this study.

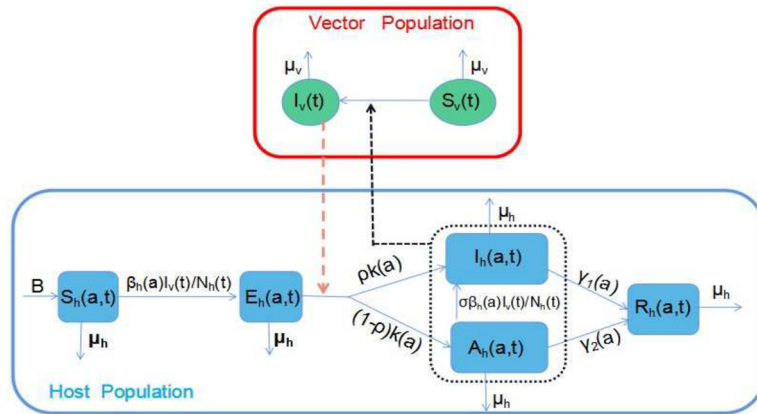


Fig. 1. Flow chart of the transitions between the host and vector compartments.

The remainder of this paper is organized as follows. In Section 2, our age-structured dengue model is described in detail. In Section 3, the basic reproduction number is derived and the stability of the disease-free steady state distribution is analyzed. Section 4 is devoted to the analysis of the existence of the endemic steady state distribution and the possible occurrence of a backward bifurcation. In Section 5, the uniform persistence of the disease is investigated. Finally, conclusions are drawn in Section 6.

2. Model formulation

We divide the human population into the susceptible, exposed, asymptomatic infectious, symptomatic infectious, and recovered individuals. Let $S_h(a, t)$, $E_h(a, t)$, $I_h(a, t)$, $A_h(a, t)$ and $R_h(a, t)$ denote, respectively, the densities of the susceptible, exposed, asymptomatic infectious, symptomatic infectious, and recovered individuals at age a and time t . We also divide the vector population into the susceptible and infectious mosquitoes.

Susceptible human individuals become exposed through bites by infected mosquitoes. We represent this process by a standard incidence $\beta_h(a)I_v(t)\frac{S_h(t)}{N_h(t)}$, where $\beta_h(a)$ is the age-dependent contact rate between infectious mosquitoes and susceptible humans and $N_h(t)$ is the total number of humans at time t . After an age-dependent latent period, exposed individuals become infectious at a rate $k(a)$. We assume that a fraction $1 - \rho$ ($0 < \rho < 1$) of these exposed individuals enter the asymptomatic infectious class. Symptomatic and asymptomatic individuals recover at rates $\gamma_1(a)$ and $\gamma_2(a)$, respectively. Meanwhile, asymptomatic individuals can be reinfected through mosquito bites with a contact rate $\sigma\beta_h(a)$ and become symptomatic infectious, where the factor σ is introduced to distinguish the contact rates between primary and secondary infections. In addition, we let μ_h denote the natural death rate of human hosts. We assume that the influx rate of the susceptible human population is a constant B .

Following prior modeling studies (e.g., [7]), we assume that infected mosquitoes never recover and carry the dengue virus in their entire lifespans. For simplicity, we re-scale the total number of mosquitoes to 1. Let $I_v(t)$ denote the portion of the infectious mosquitoes at time t . Then the portion of the susceptible mosquitoes at time t is $1 - I_v(t)$. Susceptible mosquitoes become infected through biting asymptotically or symptomatically infectious human hosts. Let $\beta_v(a)$ be the transmission rate between an infectious human at age a and a susceptible mosquito. Then the force of infection from the infected hosts to susceptible mosquitoes is $\frac{1}{N_h(t)} \int_0^\infty \beta_v(a)(I_h(a, t) + \sigma A_h(a, t))da$. We additionally denote the natural death rate of mosquitoes by μ_v .

The flow diagram for the interaction between human hosts and mosquitoes is illustrated in Fig. 1. Our age-structured model for dengue transmission dynamics is given by the following equations:

Table 1
Parameter meanings in system (2.2).

Parameter	Biological Meaning
B	The constant influx rate of the human hosts
$k(a)^{-1}$	The age-specific latent period of the human hosts
μ_h	The natural death rate of the human hosts
μ_v	The natural death rate of the mosquitoes
$\gamma_1(a)$	The age-specific recovery rate of symptomatic infectious hosts
$\gamma_2(a)$	The age-specific recovery rate of asymptomatic infectious hosts
$\beta_v(a)$	The age-specific transmission rate from infectious hosts to susceptible mosquitoes
$\beta_h(a)$	The age-specific transmission rate from infected mosquitoes to susceptible hosts
ρ	The probability of exposed hosts transferring to symptomatic infectious hosts
σ	The factor characterizing the intensity of reinfection of asymptomatic infectious hosts

$$\begin{aligned}
\frac{\partial S_h(a, t)}{\partial a} + \frac{\partial S_h(a, t)}{\partial t} &= -\beta_h(a)I_v(t)\frac{S_h(a, t)}{N_h(t)} - \mu_h S_h(a, t), \\
\frac{\partial E_h(a, t)}{\partial a} + \frac{\partial E_h(a, t)}{\partial t} &= \beta_h(a)I_v(t)\frac{S_h(a, t)}{N_h(t)} - (\mu_h + k(a))E_h(a, t), \\
\frac{\partial I_h(a, t)}{\partial a} + \frac{\partial I_h(a, t)}{\partial t} &= \rho k(a)E_h(a, t) + \sigma\beta_h(a)I_v(t)\frac{A_h(a, t)}{N_h(t)} - (\mu_h + \gamma_1(a))I_h(a, t), \\
\frac{\partial A_h(a, t)}{\partial a} + \frac{\partial A_h(a, t)}{\partial t} &= (1 - \rho)k(a)E_h(a, t) - \sigma\beta_h(a)I_v(t)\frac{A_h(a, t)}{N_h(t)} - (\mu_h + \gamma_2(a))A_h(a, t), \\
\frac{\partial R_h(a, t)}{\partial a} + \frac{\partial R_h(a, t)}{\partial t} &= \gamma_1(a)I_h(a, t) + \gamma_2(a)A_h(a, t) - \mu_h R_h(a, t), \\
\frac{dI_v(t)}{dt} &= (1 - I_v(t))\frac{1}{N_h(t)} \int_0^\infty \beta_v(a)(I_h(a, t) + \sigma A_h(a, t))da - \mu_v I_v(t),
\end{aligned} \tag{2.1}$$

with the boundary conditions

$$S_h(0, t) = B, \quad E_h(0, t) = A_h(0, t) = I_h(0, t) = R_h(0, t) = 0,$$

for $t \geq 0$, and the initial conditions

$$\begin{aligned}
S_h(a, 0) &= S_{h0}(a), \quad E_h(a, 0) = E_{h0}(a), \quad I_h(a, 0) = I_{h0}(a), \\
A_h(a, 0) &= A_{h0}(a), \quad R_h(a, 0) = R_{h0}(a), \quad I_v(0) = I_{v0},
\end{aligned}$$

for $a \geq 0$. The parameters in system (2.1) are described in Table 1. All these model parameters are assumed to be positive. In addition, we make the following assumptions.

Assumption 2.1.

- (1) $\beta_h(a), \beta_v(a), k(a), \gamma_1(a), \gamma_2(a) \in L_+^\infty[0, \infty)$.
- (2) For $x = \beta_h, \beta_v, k, \gamma_1, \gamma_2$, there exist a constant $a_x \in [0, \infty)$ and a positive constant $\underline{x}$ such that $x(a) > \underline{x}$ in a neighborhood of a_x .
- (3) $S_{h0}(a), E_{h0}(a), I_{h0}(a), A_{h0}(a), R_{h0}(a) \in L_+^1[0, \infty)$.

Assumption 2.1 implies that there exists a positive constant $\bar{x}$ such that

$$\text{ess. sup}_{a \in [0, \infty)} x(a) \leq \bar{x},$$

for $x = \beta_h, \beta_v, k, \gamma_1, \gamma_2$. Let $P(a, t)$ be the total density of the human hosts at age a and time t . Then we have

$$P(a, t) = S_h(a, t) + E_h(a, t) + I_h(a, t) + A_h(a, t) + R_h(a, t).$$

Furthermore, $P(a, t)$ satisfies the following initial-boundary-value problem:

$$\begin{aligned} \left(\frac{\partial}{\partial a} + \frac{\partial}{\partial t} \right) P(a, t) &= -\mu_h P(a, t), \\ P(0, t) &= B, \\ P(a, 0) &= P_0(a) \triangleq S_{h0}(a) + E_{h0}(a) + I_{h0}(a) + A_{h0}(a) + R_{h0}(a). \end{aligned}$$

Using the method of characteristic lines, we obtain

$$P(a, t) = \begin{cases} Be^{-\mu_h a}, & a \leq t, \\ P_0(a-t)e^{-\mu_h t}, & t < a, \end{cases}$$

and as $t \rightarrow \infty$,

$$P(a, t) \rightarrow Be^{-\mu_h a} \triangleq P_\infty(a).$$

Thus, $N_h(t)$ approaches a constant value

$$\int_0^\infty P_\infty(a) da = \frac{B}{\mu_h} \triangleq N_h^*.$$

Without loss of generality, we assume that $P(a, t) = P_\infty(a)$. Note that we do not need to consider the equation for $R_h(a, t)$ since it is decoupled from the other equations in the system. We take the following variable conversions:

$$s_h(a, t) = \frac{S_h(a, t)}{P_\infty(a)}, \quad e_h(a, t) = \frac{E_h(a, t)}{P_\infty(a)}, \quad i_h(a, t) = \frac{I_h(a, t)}{P_\infty(a)}, \quad z_h(a, t) = \frac{A_h(a, t)}{P_\infty(a)}.$$

Then we have $s_h(a, t) + e_h(a, t) + i_h(a, t) + z_h(a, t) \leq 1$. Thereafter, we consider the following normalized system:

$$\begin{aligned} \frac{\partial s_h(a, t)}{\partial a} + \frac{\partial s_h(a, t)}{\partial t} &= -\beta_h(a) I_v(t) \frac{s_h(a, t)}{N_h^*}, \\ \frac{\partial e_h(a, t)}{\partial a} + \frac{\partial e_h(a, t)}{\partial t} &= \beta_h(a) I_v(t) \frac{s_h(a, t)}{N_h^*} - k(a) e_h(a, t), \\ \frac{\partial i_h(a, t)}{\partial a} + \frac{\partial i_h(a, t)}{\partial t} &= \rho k(a) e_h(a, t) + \sigma \beta_h(a) I_v(t) \frac{z_h(a, t)}{N_h^*} - \gamma_1(a) i_h(a, t), \\ \frac{\partial z_h(a, t)}{\partial a} + \frac{\partial z_h(a, t)}{\partial t} &= (1 - \rho) k(a) e_h(a, t) - \sigma \beta_h(a) I_v(t) \frac{z_h(a, t)}{N_h^*} - \gamma_2(a) z_h(a, t), \\ \frac{dI_v(t)}{dt} &= (1 - I_v(t)) \frac{1}{N_h^*} \int_0^\infty \beta_v(a) P_\infty(a) (i_h(a, t) + \sigma z_h(a, t)) da - \mu_v I_v(t), \end{aligned} \tag{2.2}$$

equipped with the following boundary and initial conditions:

$$s_h(0, t) = 1, \quad e_h(0, t) = z_h(0, t) = i_h(0, t) = 0,$$

and

$$\begin{aligned} s_h(a, 0) = s_{h0}(a) &\triangleq \frac{S_{h0}(a)}{P_\infty(a)}, \quad e_h(a, 0) = e_{h0}(a) \triangleq \frac{E_{h0}(a)}{P_\infty(a)}, \quad i_h(a, 0) = i_{h0}(a) \triangleq \frac{I_{h0}(a)}{P_\infty(a)}, \\ z_h(a, 0) = z_{h0}(a) &\triangleq \frac{A_{h0}(a)}{P_\infty(a)}, \quad I_v(0) = I_{v0}. \end{aligned}$$

Let us define

$$X = L^1([0, \infty), \mathbb{R}), \quad Y = X \times X \times X \times X \times \mathbb{R},$$

which are equipped with the following norms

$$\begin{aligned} \|\phi\|_X &= \int_0^\infty \phi(a) da, \quad \phi \in X, \\ \|\bar{\phi}\|_Y &= \sum_{i=1}^4 \|\bar{\phi}_i\|_X + |\bar{\phi}_5|, \quad \bar{\phi} = (\bar{\phi}_1, \bar{\phi}_2, \bar{\phi}_3, \bar{\phi}_4, \bar{\phi}_5)^T \in Y, \end{aligned}$$

where the superscript T denotes the transpose of a vector. Furthermore, the positive cones of X and Y are denoted by X_+ and Y_+ , respectively. Then we define the state space for system (2.2) by

$$\Omega = \left\{ (s_h, e_h, i_h, z_h, I_v)^T \in Y_+ : 0 \leq s_h + e_h + i_h + z_h \leq 1, I_v \leq \frac{\bar{\beta}_v(1+\sigma)}{\beta_v(1+\sigma) + \mu_v} \right\}.$$

To prove the existence of a solution to system (2.2), we first define a linear operator $A : D(A) \subset Y \rightarrow Y$ by

$$A\phi(a) = \begin{pmatrix} -\frac{d}{da}\phi_1(a) \\ -\frac{d}{da}\phi_2(a) \\ -\frac{d}{da}\phi_3(a) \\ -\frac{d}{da}\phi_4(a) \\ -\mu_v\phi_5 \end{pmatrix},$$

$$\begin{aligned} D(A) &= \left\{ \phi \in Y : \phi_j \text{ is absolutely continuous and } \phi'_j \text{ (} j = 1, 2, 3, 4 \text{)} \in L^1([0, \infty), \mathbb{R}), \right. \\ &\quad \left. \phi_1(0) = 1, \phi_2(0) = \phi_3(0) = \phi_4(0) = 0 \right\}, \end{aligned}$$

and a nonlinear operator $F : \Omega \subset Y \rightarrow Y$ by

$$F\phi(a) = \begin{pmatrix} -\beta_h(a)\phi_5 \frac{\phi_1(a)}{N_h^*} \\ \beta_h(a)\phi_5 \frac{\phi_1(a)}{N_h^*} - k(a)\phi_2(a) \\ \rho k(a)\phi_2(a) + \sigma\beta_h(a)\phi_5 \frac{\phi_4(a)}{N_h^*} - \gamma_1(a)\phi_3(a) \\ (1 - \rho)k(a)\phi_2(a) - \sigma\beta_h(a)\phi_5 \frac{\phi_4(a)}{N_h^*} - \gamma_2(a)\phi_4(a) \\ (1 - \phi_5) \frac{1}{N_h^*} \int_0^\infty \beta_v(a)P_\infty(a)(\phi_3(a) + \sigma\phi_4(a))da \end{pmatrix}, \quad \phi \in \Omega.$$

Setting $u = (s_h, e_h, i_h, z_h, I_v)^T \in Y$, we can rewrite system (2.2) as the following abstract Cauchy problem in Ω :

$$\frac{d}{dt}u(t) = (A - cId)u(t) + (cId + F)u(t), \quad u(0) = u_0 \in \Omega, \quad (2.3)$$

where c is a positive constant and Id denotes the identity operator. Note that the operator $A - cId$ is an infinitesimal generator of a C_0 -semigroup $\{e^{t(A-cId)}\}_{t \geq 0} : Y \rightarrow Y$ defined by

$$e^{t(A-cId)}\phi(a) = \begin{cases} 0, & a \leq t, \\ e^{-ct}\phi(a-t), & t < a. \end{cases}$$

Obviously, $e^{t(A-cId)}(\Omega) \subset \Omega$ for all $t \geq 0$. Meanwhile, we have the following result.

Lemma 2.1. *The nonlinear operator $F : \Omega \rightarrow Y$ is Lipschitz continuous and there exists a positive constant c large enough such that*

$$\int_0^t e^{-c(t-s)}(cId + F)(\Omega)ds \subset \Omega.$$

Proof. Lipschitz continuity of F is obvious. For any $(\phi_1, \phi_2, \phi_3, \phi_4, \phi_5)^T \in \Omega$, set

$$\begin{pmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \\ \hat{\phi}_3 \\ \hat{\phi}_4 \\ \hat{\phi}_5 \end{pmatrix} = \int_0^t e^{-c(t-s)}(cId + F) \begin{pmatrix} \phi_1(s) \\ \phi_2(s) \\ \phi_3(s) \\ \phi_4(s) \\ \phi_5 \end{pmatrix} ds.$$

Then we obtain

$$\begin{aligned} \hat{\phi}_1 + \hat{\phi}_2 + \hat{\phi}_3 + \hat{\phi}_4 &= \int_0^t e^{-c(t-s)}[c(\phi_1(s) + \phi_2(s) + \phi_3(s) + \phi_4(s)) - \gamma_1(s)\phi_3(s) - \gamma_2(s)\phi_4(s)]ds \\ &\leq c \int_0^t e^{-c(t-s)}ds = 1 - e^{-ct} \leq 1, \end{aligned}$$

which shows that

$$\hat{\phi}_1 + \hat{\phi}_2 + \hat{\phi}_3 + \hat{\phi}_4 \leq 1$$

for any positive constant $c \in \mathbb{R}$. Also notice that

$$\begin{aligned} \hat{\phi}_5 &= \int_0^t e^{-(c+\mu_v+(1+\sigma)\bar{\beta}_v)(t-s)} \left[c\phi_5(s) + (1 - \phi_5(s)) \frac{1}{N_h^*} \int_0^\infty \beta_v(a)P_\infty(a) (\phi_3(a) + \sigma\phi_4(a)) da \right] ds \\ &\leq (1 + \sigma)\bar{\beta}_v \left(\frac{c}{\mu_v + (1+\sigma)\bar{\beta}_v} + 1 \right) \int_0^t e^{-(c+\mu_v+(1+\sigma)\bar{\beta}_v)(t-s)} ds \end{aligned}$$

$$\leq \frac{(1+\sigma)\bar{\beta}_v}{\mu_v + (1+\sigma)\bar{\beta}_v} (1 - e^{-(c+\mu_v+(1+\sigma)\bar{\beta}_v)t}) \leq \frac{(1+\sigma)\bar{\beta}_v}{\mu_v + (1+\sigma)\bar{\beta}_v}.$$

Hence, we can choose $c > \mu_v + (1+\sigma)\bar{\beta}_v$ such that $\hat{\phi} \in \Omega$. The proof is completed. $\square$

Employing the variation of constants formula, we know that the Cauchy problem (2.3) has a unique mild solution given by

$$u(t) = e^{At}e^{-ct}u_0 + \int_0^t e^{-c(t-s)}e^{A(t-s)}(cId + F)u(s)ds.$$

Using the iterative method proposed by Busenberg et al. [4], together with Lemma 2.1, we demonstrate that the Cauchy problem (2.3) has a mild solution in Ω . Besides, Assumption 2.1 yields that the nonlinear operator F is Fréchet differentiable. This implies that the Cauchy problem (2.3) has a classical solution. From the invariance of Ω , the Cauchy problem (2.3) generates a semiflow $U(t) : \Omega \rightarrow \Omega$:

$$U[u_0](t) = (s_h(\cdot, t), e_h(\cdot, t), i_h(\cdot, t), z_h(\cdot, t), I_v(t))^T, \quad u_0 \in \Omega. \quad (2.4)$$

In summary, we have the proposition below regarding the properties of $U(t)$.

Proposition 2.2. *Let $\{U(t)\}_{t \geq 0}$ be defined in (2.4). Then the following results hold:*

- (1) *The Cauchy problem (2.3) has a unique mild solution $U(t)u_0$.*
- (2) *The set Ω is positively invariant with respect to the semiflow $U(t)$.*
- (3) *If $u_0 \in \Omega$, then $U(t)u_0$ is a classical solution of the Cauchy problem (2.3).*

In the remainder of this article, we only consider the solution of the system (2.2) with initial conditions in the set Ω .

3. Disease-free steady state and basic reproduction number

Let $E^0 = (s_h^0(a), e_h^0(a), i_h^0(a), z_h^0(a), I_v^0)$ be the disease-free steady state of system (2.2). It is easy to see that

$$s_h^0(a) = 1, \quad e_h^0(a) = i_h^0(a) = z_h^0(a) = I_v^0 = 0.$$

To study the local stability of the disease-free steady state distribution, we introduce the change of variables by

$$s_h(a, t) = s_h^0(a) + \bar{s}_h(a, t), \quad e_h(a, t) = \bar{e}_h(a, t), \quad i_h(a, t) = \bar{i}_h(a, t), \quad z_h(a, t) = \bar{z}_h(a, t), \quad I_v(t) = \bar{I}_v(t),$$

where $\bar{s}_h(a, t)$, $\bar{e}_h(a, t)$, $\bar{i}_h(a, t)$, $\bar{z}_h(a, t)$ and $\bar{I}_v(t)$ are corresponding perturbation terms. Substituting the expressions above into (2.2) and dropping terms of order higher than two, we obtain the following PDE system:

$$\begin{aligned} \frac{\partial \bar{s}_h(a, t)}{\partial a} + \frac{\partial \bar{s}_h(a, t)}{\partial t} &= -\beta_h(a)\bar{I}_v(t)\frac{1}{N_h^*}, \\ \frac{\partial \bar{e}_h(a, t)}{\partial a} + \frac{\partial \bar{e}_h(a, t)}{\partial t} &= \beta_h(a)\bar{I}_v(t)\frac{1}{N_h^*} - k(a)\bar{e}_h(a, t), \end{aligned}$$

$$\begin{aligned}
\frac{\partial \bar{i}_h(a, t)}{\partial a} + \frac{\partial \bar{i}_h(a, t)}{\partial t} &= \rho k(a) \bar{e}_h(a, t) - \gamma_1(a) \bar{i}_h(a, t), \\
\frac{\partial \bar{z}_h(a, t)}{\partial a} + \frac{\partial \bar{z}_h(a, t)}{\partial t} &= (1 - \rho) k(a) \bar{e}_h(a, t) - \gamma_2(a) \bar{z}_h(a, t), \\
\frac{d \bar{I}_v(t)}{dt} &= \frac{1}{N_h^*} \int_0^\infty \beta_v(a) P_\infty(a) (\bar{i}_h(a, t) + \sigma \bar{z}_h(a, t)) da - \mu_v \bar{I}_v(t),
\end{aligned} \tag{3.1}$$

with the boundary conditions

$$\bar{s}_h(0, t) = \bar{e}_h(0, t) = \bar{i}_h(0, t) = \bar{z}_h(0, t) = 0.$$

We consider perturbations in a separable form given by

$$\bar{s}_h(a, t) = s_h(a) e^{\lambda t}, \quad \bar{e}_h(a, t) = e_h(a) e^{\lambda t}, \quad \bar{i}_h(a, t) = i_h(a) e^{\lambda t}, \quad \bar{z}_h(a, t) = z_h(a) e^{\lambda t}, \quad \bar{I}_v(t) = I_v e^{\lambda t},$$

where I_v is a constant and $s_h(0) = e_h(0) = i_h(0) = z_h(0) = 0$. Substituting them into system (3.1), we can obtain the following results after some algebraic manipulation,

$$\begin{aligned}
s_h(a) &= -\frac{I_v}{N_h^*} \int_0^a \beta_h(\tau) e^{-\lambda(a-\tau)} d\tau, \\
e_h(a) &= \frac{I_v}{N_h^*} \int_0^a \beta_h(\theta) e^{-\lambda(a-\theta)} e^{-\int_\theta^a k(u) du} d\theta, \\
i_h(a) &= \frac{\rho I_v}{N_h^*} \int_0^a \int_0^\xi k(\xi) \beta_h(\theta) e^{-\lambda(a-\theta)} e^{-\int_\theta^\xi k(u) du} e^{-\int_\xi^a \gamma_1(u) du} d\theta d\xi, \\
z_h(a) &= \frac{(1-\rho) I_v}{N_h^*} \int_0^a \int_0^\xi k(\xi) \beta_h(\theta) e^{-\lambda(a-\theta)} e^{-\int_\theta^\xi k(u) du} e^{-\int_\xi^a \gamma_2(u) du} d\theta d\xi, \\
I_v &= \frac{1}{N_h^*(\lambda + \mu_v)} \int_0^\infty \beta_v(a) P_\infty(a) [i_h(a) + \sigma z_h(a)] da.
\end{aligned} \tag{3.2}$$

Assume $I_v \neq 0$. Substituting $i_h(a)$ and $z_h(a)$ into the equation for I_v , we obtain that λ is the solution of the following characteristic equation:

$$\begin{aligned}
1 &= \frac{1}{N_h^*(\lambda + \mu_v)} \int_0^\infty \beta_v(a) P_\infty(a) \left(\frac{\rho}{N_h^*} \int_0^a \int_0^\xi k(\xi) \beta_h(\theta) e^{-\lambda(a-\theta)} e^{-\int_\theta^\xi k(u) du} e^{-\int_\xi^a \gamma_1(u) du} d\theta d\xi \right. \\
&\quad \left. + \frac{\sigma(1-\rho)}{N_h^*} \int_0^a \int_0^\xi k(\xi) \beta_h(\theta) e^{-\lambda(a-\theta)} e^{-\int_\theta^\xi k(u) du} e^{-\int_\xi^a \gamma_2(u) du} d\theta d\xi \right) da \triangleq G(\lambda).
\end{aligned} \tag{3.3}$$

The basic reproduction number is then defined as

$$\mathcal{R}_0 = G(0) = \frac{1}{\mu_v N_h^*} \int_0^\infty \beta_v(a) P_\infty(a) \left(\frac{\rho}{N_h^*} \int_0^a \int_0^\xi k(\xi) \beta_h(\theta) e^{-\int_\theta^\xi k(u) du} e^{-\int_\xi^a \gamma_1(u) du} d\theta d\xi \right.$$

$$+ \frac{\sigma(1-\rho)}{N_h^*} \int_0^a \int_0^\xi k(\xi) \beta_h(\theta) e^{-\int_\theta^\xi k(u) du} e^{-\int_\xi^a \gamma_2(u) du} d\theta d\xi \Big) da.$$

Biologically, the basic reproduction number $\mathcal{R}_0$ can be interpreted as the average number of secondary cases that one infectious mosquito produces during its lifespan in a human population. From the expression of $\mathcal{R}_0$, we see that it consists of two parts, one for the transmission through the symptomatic hosts and the other for the transmission through the asymptomatic hosts. In the following, we will use $\mathcal{R}_0$ to analyze the stability of the disease-free steady state distribution.

Theorem 3.1. *The disease-free steady state E^0 of system (2.2) is locally asymptotically stable if $\mathcal{R}_0 < 1$, and unstable if $\mathcal{R}_0 > 1$.*

Proof. It is easy to see that $G(\lambda)$ is a strictly decreasing function in $[0, \infty)$ and $\lim_{\lambda \rightarrow \infty} G(\lambda) = 0$. Then, if $\mathcal{R}_0 > 1$, the characteristic equation has one positive root, which shows that the disease-free steady state is unstable. Next, we show that if $\mathcal{R}_0 < 1$, then all solutions of the equation $G(\lambda) = 1$ have negative real parts. Let $\lambda_0 = p + iq$ be a root in a complex form. Assume $p \geq 0$. Substituting it into the characteristic equation, we obtain the following:

$$\begin{aligned} 1 = |G(\lambda_0)| &\leq \frac{1}{N_h^* \sqrt{(p + \mu_v)^2 + q^2}} \int_0^\infty \beta_v(a) P_\infty(a) \left(\frac{\rho}{N_h^*} \int_0^a \int_0^\xi k(\xi) \beta_h(\theta) e^{-p(a-\theta)} e^{-\int_\theta^\xi k(u) du} \right. \\ &\quad \left. e^{-\int_\xi^a \gamma_1(u) du} d\theta d\xi + \frac{\sigma(1-\rho)}{N_h^*} \int_0^a \int_0^\xi k(\xi) \beta_h(\theta) e^{-p(a-\theta)} e^{-\int_\theta^\xi k(u) du} e^{-\int_\xi^a \gamma_2(u) du} d\theta d\xi \right) da \\ &\leq \frac{1}{N_h^* (p + \mu_v)} \int_0^\infty \beta_v(a) P_\infty(a) \left(\frac{\rho}{N_h^*} \int_0^a \int_0^\xi k(\xi) \beta_h(\theta) e^{-p(a-\theta)} e^{-\int_\theta^\xi k(u) du} e^{-\int_\xi^a \gamma_1(u) du} d\theta d\xi \right. \\ &\quad \left. + \frac{\sigma(1-\rho)}{N_h^*} \int_0^a \int_0^\xi k(\xi) \beta_h(\theta) e^{-p(a-\theta)} e^{-\int_\theta^\xi k(u) du} e^{-\int_\xi^a \gamma_2(u) du} d\theta d\xi \right) da \\ &= G(p) \leq G(0) = \mathcal{R}_0 < 1, \end{aligned}$$

which is a contradiction. Hence, the disease-free steady state distribution is locally asymptotically stable. $\square$

Let

$$\begin{aligned} \Gamma(a, \theta, \varepsilon, \xi) &= \beta_h(\varepsilon) k(\xi) \beta_h(\theta) e^{-\int_\theta^\xi k(\phi) d\phi} e^{-\int_\xi^\varepsilon \gamma_2(\phi) d\phi} e^{-\int_\varepsilon^a \gamma_1(\phi) d\phi}, \\ \varsigma &= \frac{\sigma(1-\rho)}{\mu_v N_h^{*3}} \int_0^\infty \int_0^a \int_0^\varepsilon \int_0^\xi \beta_v(a) P_\infty(a) \Gamma(a, \theta, \varepsilon, \xi) d\theta d\xi d\varepsilon da. \end{aligned}$$

Then we can also establish the following result about the global stability of the disease-free steady state.

Theorem 3.2. *If $\mathcal{R}_0 + \varsigma < 1$, the disease-free steady state E^0 of system (2.2) is globally asymptotically stable.*

Proof. For system (2.2), by the method of characteristic lines, we have

$$s_h(a, t) = e^{-\int_0^a \frac{\beta_h(\tau) I_v(t-a+\tau)}{N_h^*} d\tau},$$

$$\begin{aligned}
e_h(a, t) &= \int_0^a \frac{\beta_h(\theta) I_v(t-a+\theta)}{N_h^*} e^{-\int_0^\theta \frac{\beta_h(\tau) I_v(t-a+\tau)}{N_h^*} d\tau} e^{-\int_\theta^a k(\phi) d\phi} d\theta, \\
z_h(a, t) &= (1-\rho) \int_0^a \int_0^\xi k(\xi) \frac{\beta_h(\theta) I_v(t-a+\theta)}{N_h^*} e^{-\int_0^\theta \frac{\beta_h(\tau) I_v(t-a+\tau)}{N_h^*} d\tau} e^{-\int_\theta^\xi k(\phi) d\phi} \\
&\quad e^{-\int_\xi^a [\frac{\sigma \beta_h(\phi) I_v(t-a+\phi)}{N_h^*} + \gamma_2(\phi)] d\phi} d\theta d\xi, \\
i_h(a, t) &= \int_0^a \left(\rho k(\varepsilon) \int_0^\varepsilon \frac{\beta_h(\theta) I_v(t-a+\theta)}{N_h^*} e^{-\int_0^\theta \frac{\beta_h(\tau) I_v(t-a+\tau)}{N_h^*} d\tau} e^{-\int_\theta^\varepsilon k(\phi) d\phi} d\theta \right. \\
&\quad \left. + \frac{\sigma \beta_h(\varepsilon) I_v(t-a+\varepsilon)}{N_h^*} (1-\rho) \int_0^\varepsilon \int_0^\xi k(\xi) \frac{\beta_h(\theta) I_v(t-a+\theta)}{N_h^*} e^{-\int_0^\theta \frac{\beta_h(\tau) I_v(t-a+\tau)}{N_h^*} d\tau} \right. \\
&\quad \left. e^{-\int_\theta^\xi k(\phi) d\phi} e^{-\int_\xi^\varepsilon [\frac{\sigma \beta_h(\phi) I_v(t-a+\phi)}{N_h^*} + \gamma_2(\phi)] d\phi} d\theta d\xi \right) e^{-\int_\varepsilon^a \gamma_1(\phi) d\phi} d\varepsilon,
\end{aligned}$$

for $t \geq a$. Then we can obtain

$$\begin{aligned}
i_h(a, t) + \sigma z_h(a, t) &= \frac{\rho}{N_h^*} \int_0^a \int_0^\xi k(\xi) \beta_h(\theta) I_v(t-a+\theta) e^{-\int_0^\theta \frac{\beta_h(\tau) I_v(t-a+\tau)}{N_h^*} d\tau} e^{-\int_\theta^\xi k(\phi) d\phi} \\
&\quad e^{-\int_\xi^a \gamma_1(\phi) d\phi} d\theta d\xi + (1-\rho) \int_0^a \int_0^\varepsilon \int_0^\xi \frac{\sigma \beta_h(\varepsilon) I_v(t-a+\varepsilon)}{N_h^{*2}} k(\xi) \beta_h(\theta) I_v(t-a+\theta) \\
&\quad e^{-\int_0^\theta \frac{\beta_h(\tau) I_v(t-a+\tau)}{N_h^*} d\tau} e^{-\int_\theta^\xi k(\phi) d\phi} e^{-\int_\xi^\varepsilon [\frac{\sigma \beta_h(\phi) I_v(t-a+\phi)}{N_h^*} + \gamma_2(\phi)] d\phi} e^{-\int_\varepsilon^a \gamma_1(\phi) d\phi} d\theta d\xi d\varepsilon \\
&\quad + \frac{\sigma(1-\rho)}{N_h^*} \int_0^a \int_0^\xi k(\xi) \beta_h(\theta) I_v(t-a+\theta) e^{-\int_0^\theta \frac{\beta_h(\tau) I_v(t-a+\tau)}{N_h^*} d\tau} e^{-\int_\theta^\xi k(\phi) d\phi} \\
&\quad e^{-\int_\xi^a [\frac{\sigma \beta_h(\phi) I_v(t-a+\phi)}{N_h^*} + \gamma_2(\phi)] d\phi} d\theta d\xi \\
&\leq \frac{\rho}{N_h^*} \int_0^a \int_0^\xi k(\xi) \beta_h(\theta) I_v(t-a+\theta) e^{-\int_\theta^\xi k(\phi) d\phi} e^{-\int_\xi^a \gamma_1(\phi) d\phi} d\theta d\xi \\
&\quad + \frac{\sigma(1-\rho)}{N_h^*} \int_0^a \int_0^\xi k(\xi) \beta_h(\theta) I_v(t-a+\theta) e^{-\int_\theta^\xi k(\phi) d\phi} e^{-\int_\xi^a \gamma_2(\phi) d\phi} d\theta d\xi \\
&\quad + \frac{\sigma(1-\rho)}{N_h^{*2}} \int_0^a \int_0^\varepsilon \int_0^\xi \Gamma(a, \theta, \varepsilon, \xi) I_v(t-a+\varepsilon) d\theta d\xi d\varepsilon. \tag{3.4}
\end{aligned}$$

By Fluctuation Lemma [17], there exists a sequence $\{t_n\}$ such that

$$t_n \rightarrow \infty, \quad I_v(t_n) \rightarrow \limsup_{t \rightarrow \infty} I_v(t) \triangleq I_v^\infty, \quad \frac{dI_v(t_n)}{dt} \rightarrow 0,$$

as $n \rightarrow \infty$. Using the equation for $I_v(t)$ in system (2.2), we have

$$\mu_v I_v(t_n) + \frac{dI_v(t_n)}{dt} = (1 - I_v(t_n)) \frac{1}{N_h^*} \int_0^{t_n} \beta_v(a) P_\infty(a) (i_h(a, t_n) + \sigma z_h(a, t_n)) da + L(t_n), \quad (3.5)$$

where

$$L(t) = (1 - I_v(t)) \frac{1}{N_h^*} \int_t^\infty \beta_v(a) P_\infty(a) (i_h(a, t) + \sigma z_h(a, t)) da.$$

Taking the limit with $n \rightarrow \infty$ on both sides of the equation (3.5), and using Fatou's lemma and the fact $\lim_{t \rightarrow \infty} L(t) = 0$, we obtain

$$\begin{aligned} \mu_v I_v^\infty &= \lim_{n \rightarrow \infty} \left((1 - I_v(t_n)) \frac{1}{N_h^*} \int_0^\infty \beta_v(a) P_\infty(a) (i_h(a, t_n) + \sigma z_h(a, t_n)) da \right) \\ &= \limsup_{n \rightarrow \infty} \left((1 - I_v(t_n)) \frac{1}{N_h^*} \int_0^\infty \beta_v(a) P_\infty(a) (i_h(a, t_n) + \sigma z_h(a, t_n)) da \right) \\ &\leq \frac{1}{N_h^*} \limsup_{n \rightarrow \infty} \left((1 - I_v(t_n)) \int_0^\infty \beta_v(a) P_\infty(a) (i_h(a, t_n) + \sigma z_h(a, t_n)) da \right) \\ &\leq \frac{1}{N_h^*} (1 - I_v^\infty) \int_0^\infty \beta_v(a) P_\infty(a) \limsup_{n \rightarrow \infty} (i_h(a, t_n) + \sigma z_h(a, t_n)) da. \end{aligned}$$

Thus, we have

$$I_v^\infty \leq \frac{1}{\mu_v N_h^*} (1 - I_v^\infty) \int_0^\infty \beta_v(a) P_\infty(a) \limsup_{n \rightarrow \infty} (i_h(a, t_n) + \sigma z_h(a, t_n)) da.$$

It follows from (3.4) that

$$\begin{aligned} \limsup_{n \rightarrow \infty} [i_h(a, t_n) + \sigma z_h(a, t_n)] &\leq I_v^\infty \left(\frac{\rho}{N_h^*} \int_0^a \int_0^\xi k(\xi) \beta_h(\theta) e^{-\int_\theta^\xi k(\phi) d\phi} e^{-\int_\xi^a \gamma_1(\phi) d\phi} d\theta d\xi \right. \\ &\quad + \frac{\sigma(1-\rho)}{N_h^*} \int_0^a \int_0^\xi k(\xi) \beta_h(\theta) e^{-\int_\theta^\xi k(\phi) d\phi} e^{-\int_\xi^a \gamma_2(\phi) d\phi} d\theta d\xi \\ &\quad \left. + \frac{\sigma(1-\rho)}{N_h^{*2}} \int_0^a \int_0^\varepsilon \int_0^\xi \Gamma(a, \theta, \varepsilon, \xi) d\theta d\xi d\varepsilon \right), \end{aligned}$$

which yields

$$I_v^\infty \leq I_v^\infty (1 - I_v^\infty) \frac{1}{\mu_v N_h^*} \int_0^\infty \beta_v(a) P_\infty(a) \left(\frac{\rho}{N_h^*} \int_0^a \int_0^\xi k(\xi) \beta_h(\theta) e^{-\int_\theta^\xi k(\phi) d\phi} e^{-\int_\xi^a \gamma_1(\phi) d\phi} d\theta d\xi \right.$$

$$\begin{aligned}
& + \frac{\sigma(1-\rho)}{N_h^*} \int_0^a \int_0^\xi k(\xi) \beta_h(\theta) e^{-\int_\theta^\xi k(\phi) d\phi} e^{-\int_\xi^a \gamma_2(\phi) d\phi} d\theta d\xi \\
& + \frac{\sigma(1-\rho)}{N_h^{*2}} \int_0^a \int_0^\varepsilon \int_0^\xi \Gamma(a, \theta, \varepsilon, \xi) d\theta d\xi d\varepsilon \Big) da \\
& = I_v^\infty (1 - I_v^\infty) \left[\frac{\sigma(1-\rho)}{\mu_v N_h^{*3}} \int_0^\infty \int_0^a \int_0^\varepsilon \int_0^\xi \beta_v(a) P_\infty(a) \Gamma(a, \theta, \varepsilon, \xi) d\theta d\xi d\varepsilon da \right. \\
& + \frac{1}{\mu_v N_h^*} \int_0^\infty \beta_v(a) P_\infty(a) \left(\frac{\rho}{N_h^*} \int_0^a \int_0^\xi k(\xi) \beta_h(\theta) e^{-\int_\theta^\xi k(\phi) d\phi} e^{-\int_\xi^a \gamma_1(\phi) d\phi} d\theta d\xi \right. \\
& \left. \left. + \frac{\sigma(1-\rho)}{N_h^*} \int_0^a \int_0^\xi k(\xi) \beta_h(\theta) e^{-\int_\theta^\xi k(\phi) d\phi} e^{-\int_\xi^a \gamma_2(\phi) d\phi} d\theta d\xi \right) da \right] = I_v^\infty (1 - I_v^\infty) (\varsigma + \mathcal{R}_0).
\end{aligned}$$

Since $\mathcal{R}_0 + \varsigma < 1$, we have $I_v^\infty = 0$. This implies

$$\lim_{t \rightarrow \infty} s_h(a, t) = 1, \quad \lim_{t \rightarrow \infty} I_v(t) = \lim_{t \rightarrow \infty} e_h(a, t) = \lim_{t \rightarrow \infty} i_h(a, t) = \lim_{t \rightarrow \infty} z_h(a, t) = 0.$$

The proof is then completed. $\square$

Now, we provide a numerical example to confirm the result in Theorem 3.2, where the global stability of the disease-free steady state is guaranteed if $\mathcal{R}_0 + \varsigma < 1$. We choose constant parameters by $B = 250$, $\mu_h = 0.005$, $k = 0.6$, $\sigma = 0.007$, $\beta_h = 18$, $\gamma_1 = 0.5$, $\gamma_2 = 0.5$, $\rho = 0.5$, $\mu_v = 0.65$, $\beta_v = 8$. Then $\mathcal{R}_0 + \varsigma = 0.0044 < 1$. The numerical solutions of $e_h(a, t)$, $z_h(a, t)$, $i_h(a, t)$ and $I_v(t)$ are displayed in Fig. 2. Clearly, the solutions $e_h(a, t)$, $z_h(a, t)$, $i_h(a, t)$ and $I_v(t)$ all approach zero when t is large.

4. Endemic steady state

We have shown that the disease-free steady state distribution is unstable if $\mathcal{R}_0 > 1$. In this case, it is possible that system (2.2) has an endemic steady state, which is a time-independent solution of system (2.2). In this section, we analyze the existence of such endemic steady states.

Theorem 4.1. *There is at least one endemic steady state of system (2.2) if $\mathcal{R}_0 > 1$.*

Proof. The endemic steady state of system (2.2) satisfies the following system of equations:

$$\begin{aligned}
\frac{ds_h^*(a)}{da} &= -\beta_h(a) I_v^* \frac{s_h^*(a)}{N_h^*}, \\
\frac{de_h^*(a)}{da} &= \beta_h(a) I_v^* \frac{s_h^*(a)}{N_h^*} - k(a) e_h^*(a), \\
\frac{di_h^*(a)}{da} &= \rho k(a) e_h^*(a) + \sigma \beta_h(a) I_v^* \frac{z_h^*(a)}{N_h^*} - \gamma_1(a) i_h^*(a), \\
\frac{dz_h^*(a)}{da} &= (1 - \rho) k(a) e_h^*(a) - \sigma \beta_h(a) I_v^* \frac{z_h^*(a)}{N_h^*} - \gamma_2(a) z_h^*(a),
\end{aligned} \tag{4.1}$$

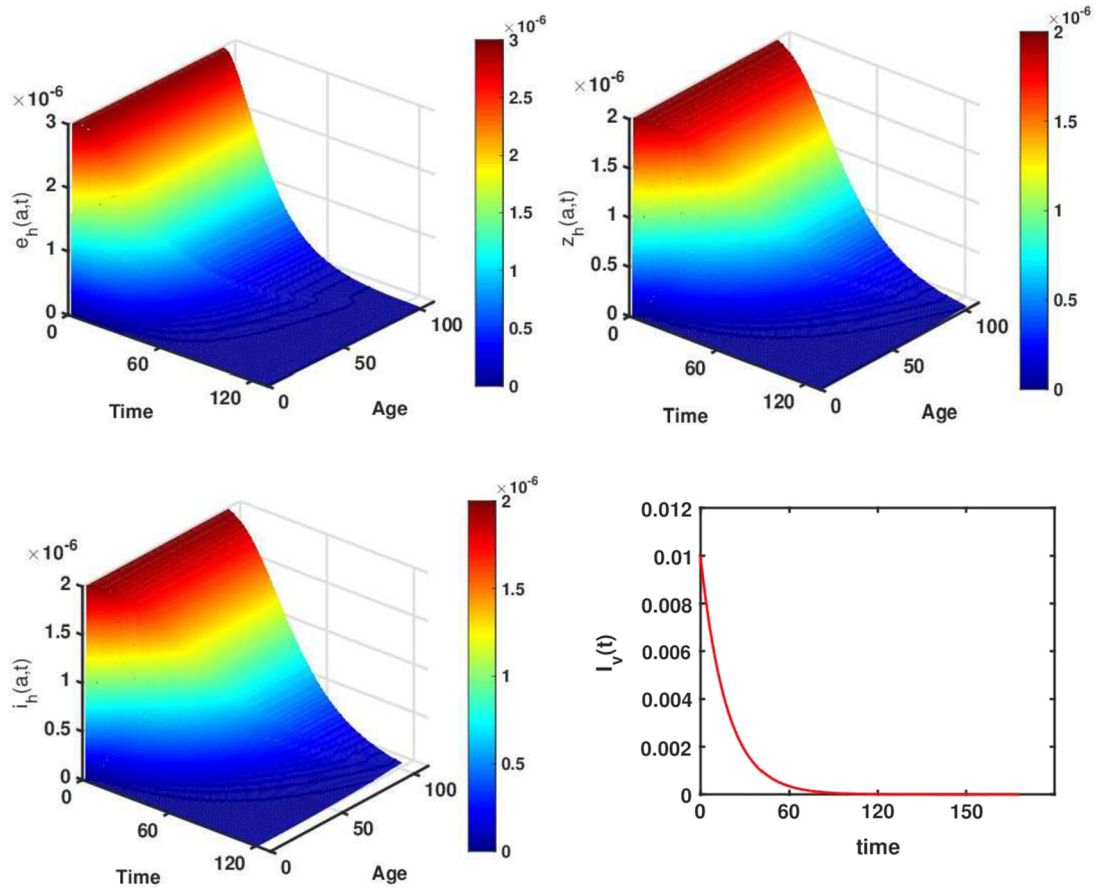


Fig. 2. A typical scenario showing the time and age series of $e_h(a, t)$, $z_h(a, t)$, $i_h(a, t)$ and $I_v(t)$ when $R_0 + \varsigma < 1$. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

$$0 = (1 - I_v^*) \frac{1}{N_h^*} \int_0^\infty \beta_v(a) P_\infty(a) (i_h^*(a) + \sigma z_h^*(a)) da - \mu_v I_v^*,$$

where $s_h^*(0) = 1$, $e_h^*(0) = i_h^*(0) = z_h^*(0) = 0$. A solution to (4.1) can be expressed as

$$\begin{aligned} s_h^*(a) &= e^{-\frac{I_v^*}{N_h^*} \int_0^a \beta_h(u) du}, \\ e_h^*(a) &= \frac{I_v^*}{N_h^*} \int_0^a \beta_h(\tau) s_h^*(\tau) e^{-\int_\tau^a k(u) du} d\tau, \\ z_h^*(a) &= (1 - \rho) \int_0^a k(\theta) e_h^*(\theta) e^{-\int_\theta^a (\frac{\sigma I_v^* \beta_h(u)}{N_h^*} + \gamma_2(u)) du} d\theta, \\ i_h^*(a) &= \int_0^a (\rho k(\xi) e_h^*(\xi) + \frac{\sigma I_v^*}{N_h^*} \beta_h(\xi) z_h^*(\xi)) e^{-\int_\xi^a \gamma_1(u) du} d\xi, \\ I_v^* &= \frac{\frac{1}{N_h^*} \int_0^\infty \beta_v(a) P_\infty(a) (i_h^*(a) + \sigma z_h^*(a)) da}{\mu_v + \frac{1}{N_h^*} \int_0^\infty \beta_v(a) P_\infty(a) (i_h^*(a) + \sigma z_h^*(a)) da} \triangleq \frac{\lambda_v}{\mu_v + \lambda_v} \in (0, 1). \end{aligned} \quad (4.2)$$

It is easy to see from (4.2) that the existence and number of these endemic steady states depend on the positive solution $I_v^* \in (0, 1)$ satisfying the system above, which in turn depends on the positive solution λ_v . Thus, we only need to analyze λ_v . After some algebraic manipulation, we obtain

$$\begin{aligned} e_h^*(a) &= \frac{I_v^*}{N_h^*} \int_0^a \beta_h(\tau) e^{-\frac{I_v^*}{N_h^*} \int_0^\tau \beta_h(u) du} e^{-\int_\tau^a k(u) du} d\tau, \\ z_h^*(a) &= (1 - \rho) \frac{I_v^*}{N_h^*} \int_0^a \int_0^\theta k(\theta) \beta_h(\tau) e^{-\frac{I_v^*}{N_h^*} \int_0^\tau \beta_h(u) du} e^{-\int_\tau^\theta k(u) du} e^{-\int_\theta^a \left(\gamma_2(u) + \frac{\sigma I_v^* \beta_h(u)}{N_h^*} \right) du} d\tau d\theta, \\ i_h^*(a) &= \rho \frac{I_v^*}{N_h^*} \int_0^a \int_0^\xi k(\xi) \beta_h(\tau) e^{-\frac{I_v^*}{N_h^*} \int_0^\tau \beta_h(u) du} e^{-\int_\tau^\xi k(u) du} e^{-\int_\xi^a \gamma_1(u) du} d\tau d\xi + \sigma (1 - \rho) \left(\frac{I_v^*}{N_h^*} \right)^2 \\ &\quad \times \int_0^a \int_0^\xi \int_0^\theta k(\theta) \beta_h(\xi) \beta_h(\tau) e^{-\int_\tau^\theta k(u) du} e^{-\frac{I_v^*}{N_h^*} \int_0^\tau \beta_h(u) du} e^{-\sigma \frac{I_v^*}{N_h^*} \int_\theta^\xi \beta_h(u) du} e^{-\int_\theta^\xi \gamma_2(u) du} \\ &\quad \times e^{-\int_\xi^a \gamma_1(u) du} d\tau d\theta d\xi. \end{aligned}$$

Substitution of $i_h^*(a)$ and $z_h^*(a)$ into the expression for λ_v yields

$$\begin{aligned} 1 &= \frac{1}{N_h^*} \int_0^\infty \beta_v(a) P_\infty(a) \left(\frac{\rho}{N_h^* (\mu_v + \lambda_v)} \int_0^a \int_0^\xi k(\xi) \beta_h(\tau) e^{-\int_\tau^\xi k(u) du} e^{-\frac{\lambda_v}{N_h^* (\mu_v + \lambda_v)} \int_0^\tau \beta_h(u) du} \right. \\ &\quad \left. e^{-\int_\xi^a \gamma_1(u) du} d\tau d\xi + \frac{(1 - \rho) \sigma \lambda_v}{(N_h^* (\mu_v + \lambda_v))^2} \int_0^a \int_0^\xi \int_0^\theta k(\theta) \beta_h(\xi) \beta_h(\tau) e^{-\int_\tau^\theta k(u) du} e^{-\frac{\lambda_v}{N_h^* (\mu_v + \lambda_v)} \int_0^\tau \beta_h(u) du} \right. \\ &\quad \times e^{-\frac{\sigma \lambda_v}{N_h^* (\mu_v + \lambda_v)} \int_\theta^\xi \beta_h(u) du} e^{-\int_\theta^\xi \gamma_2(u) du} e^{-\int_\xi^a \gamma_1(u) du} d\tau d\theta d\xi + \frac{\sigma (1 - \rho)}{N_h^* (\mu_v + \lambda_v)} \int_0^a \int_0^\theta k(\theta) \beta_h(\tau) \\ &\quad \left. \times e^{-\int_\tau^\theta k(u) du} e^{-\frac{\lambda_v}{N_h^* (\mu_v + \lambda_v)} \int_0^\tau \beta_h(u) du} e^{-\frac{\sigma \lambda_v}{N_h^* (\mu_v + \lambda_v)} \int_\theta^\tau \beta_h(u) du} e^{-\int_\theta^a \gamma_2(u) du} d\tau d\theta \right) da \triangleq F(\lambda_v). \end{aligned}$$

Then we have $\lim_{\lambda_v \rightarrow \infty} F(\lambda_v) = 0$, $F(0) = \mathcal{R}_0$. Thus, if $\mathcal{R}_0 > 1$, the equation $F(\lambda) = 1$ has at least one positive root, which suggests that there is at least one endemic steady state solution. $\square$

We again use a numerical simulation to demonstrate the result in Theorem 4.1. We choose constant parameters by $B = 200$, $\mu_h = 0.05$, $k = 0.6$, $\sigma = 0.07$, $\beta_h = 50$, $\gamma_1 = 0.5$, $\gamma_2 = 0.5$, $\rho = 0.5$, $\mu_v = 0.2$, $\beta_v = 30$, so that $\mathcal{R}_0 = 1.684 > 1$. The numerical solutions of $e_h(a, t)$, $z_h(a, t)$, $i_h(a, t)$ and $I_v(t)$ under this setting are displayed in Fig. 3. We observe that the solutions $e_h(a, t)$, $z_h(a, t)$, $i_h(a, t)$ and $I_v(t)$ all approach an endemic steady state for large t .

5. Backward bifurcation

Now we analyze the existence of endemic steady states when $\mathcal{R}_0 < 1$. To facilitate the description of the conditions, we set

$$G_1 = \int_0^\infty \frac{\beta_v(a) P_\infty(a)}{(\mu_v N_h^*)^2} \int_0^a \int_0^\theta k(\theta) \beta_h(\tau) e^{-\int_\tau^\theta k(u) du} \left[\sigma (1 - \rho) e^{-\int_\theta^a \gamma_2(u) du} \times \left(1 + \frac{1}{N_h^*} \right. \right.$$

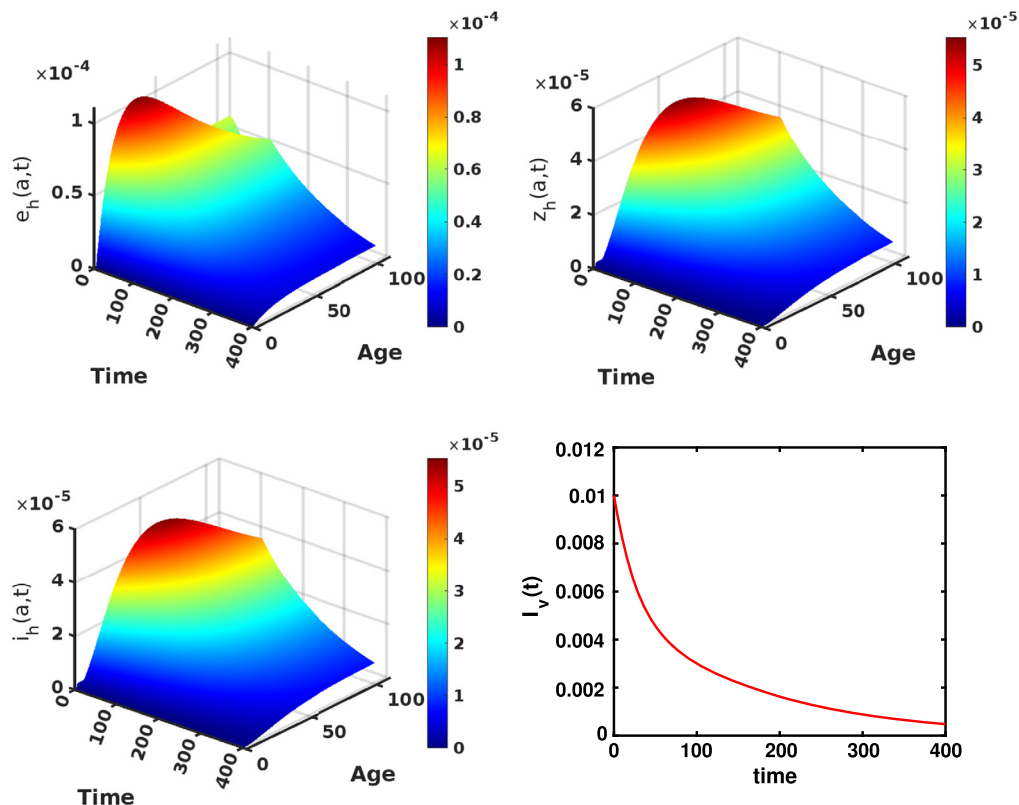


Fig. 3. A typical scenario showing the time and age series of $e_h(a, t)$, $z_h(a, t)$, $i_h(a, t)$ and $I_v(t)$ when $R_0 > 1$.

$$\begin{aligned}
 & \times \left(\int_0^\tau \beta_h(u) du + \sigma \int_\theta^a \beta_h(u) du \right) + \rho e^{-\int_\theta^a \gamma_1(u) du} \left(1 + \frac{1}{N_h^*} \int_0^\tau \beta_h(u) du \right) \Big] d\tau d\theta da, \\
 G_2 = & \int_0^\infty \frac{\sigma(1-\rho)\beta_v(a)P_\infty(a)}{\mu_v^2 N_h^{*3}} \int_0^a \int_0^\xi \int_0^\theta k(\theta)\beta_h(\xi)\beta_h(\tau) e^{-\int_\tau^\theta k(u) du} e^{-\int_\theta^\xi \gamma_2(u) du} e^{-\int_\xi^a \gamma_1(u) du} \\
 & \times d\tau d\theta d\xi da.
 \end{aligned}$$

Theorem 5.1. If $\mathcal{R}_0 < 1$, $G_1 < G_2$, and $|\mathcal{R}_0 - 1|$ is sufficiently small, then system (2.2) has at least two endemic steady states.

Proof. Consider the function with a parameter ϕ :

$$W(\lambda_v, \phi) = \phi \frac{F(\lambda_v)}{\mathcal{R}_0} - 1.$$

Then we have $W(0, 1) = 0$ and $\frac{\partial W}{\partial \lambda_v} \Big|_{(0,1)} = \frac{F'(0)}{\mathcal{R}_0} = \frac{G_2 - G_1}{\mathcal{R}_0} > 0$. It follows from the Implicit Function Theorem that the equation $W(\lambda_v, \phi) = 0$ uniquely determines a function $\lambda_v = \lambda_v(\phi)$ at a neighborhood $\Omega_0 = \{(\lambda_v, \phi) \mid -\delta_1 < \lambda_v < \delta_1, 1 - \delta_2 < \phi < 1 + \delta_2\}$ of the point $(0, 1)$, which satisfies

$$\frac{d\lambda_v(\phi)}{d\phi} \Big|_{\phi=1} = -\frac{W_\phi(0, 1)}{W_{\lambda_v}(0, 1)} = -\frac{\mathcal{R}_0}{G_2 - G_1} < 0.$$

Since $\lambda_v(1) = 0$, there exists a small $\epsilon > 0$ ($\epsilon < \delta_2$) such that $\lambda_v(\phi) > 0$ for $\phi \in (1 - \epsilon, 1)$. Clearly, $W(\lambda_v(\phi), \phi) \equiv 0$. Considering $|\mathcal{R}_0 - 1|$ sufficiently small, we confine ourselves to $\mathcal{R}_0 \in (1 - \epsilon/2, 1)$. Then we have $F(\lambda_v(\phi_0)) = \frac{\mathcal{R}_0}{\phi_0} > 1$ for any $\phi_0 \in (1 - \epsilon, 1 - \epsilon/2)$. Also noticing $\lim_{\lambda_v \rightarrow \infty} F(\lambda_v) = 0$, $F(0) = \mathcal{R}_0 < 1$, we see that the equation $F(\lambda_v) = 1$ has at least two positive roots. The proof is completed. $\square$

Next, we explore conditions that ensure the existence of exactly two endemic steady states when $\mathcal{R}_0 < 1$, which would imply a backward bifurcation. For simplicity, it is assumed that all the parameters are constants. From the above analysis, we see that if $G(0) = F(0) = \mathcal{R}_0 < 1$, there are at least two endemic steady states. Specially, if $\beta_h, \beta_v, \mu_h, \mu_v, \gamma_1, \gamma_2$ and k are all constants, we have

$$\begin{aligned} s_h^*(a) &= e^{-\frac{\beta_h \mu_h I_v^*}{B} a}, \\ e_h^*(a) &= \frac{\beta_h \mu_h I_v^*}{Bk - \beta_h \mu_h I_v^*} \left(e^{-\frac{\beta_h \mu_h I_v^*}{B} a} - e^{-ka} \right), \\ z_h^*(a) &= \frac{k(1 - \rho) \beta_h \mu_h I_v^*}{Bk - \beta_h \mu_h I_v^*} \left(\frac{1}{k - \gamma_2 - \frac{\sigma \beta_h \mu_h I_v^*}{B}} \left(e^{-ka} - e^{-(\gamma_2 + \frac{\sigma \beta_h \mu_h I_v^*}{B})a} \right) + \frac{1}{\gamma_2 + \frac{\sigma \beta_h \mu_h I_v^*}{B} - \frac{\beta_h \mu_h I_v^*}{B}} \right. \\ &\quad \left. \times \left(e^{-\frac{\beta_h \mu_h I_v^*}{B} a} - e^{-(\gamma_2 + \frac{\sigma \beta_h \mu_h I_v^*}{B})a} \right) \right), \\ i_h^*(a) &= \frac{k \rho \beta_h \mu_h I_v^*}{Bk - \beta_h \mu_h I_v^*} \left[\frac{1}{k - \gamma_1} (e^{-ka} - e^{-\gamma_1 a}) - \frac{1}{\frac{\beta_h \mu_h I_v^*}{B} - \gamma_1} \left(e^{-\frac{\beta_h \mu_h I_v^*}{B} a} - e^{-\gamma_1 a} \right) \right] \\ &\quad + \frac{k \sigma (1 - \rho) (\beta_h \mu_h I_v^*)^2}{B^2 k - B \beta_h \mu_h I_v^*} \left\{ \frac{1}{k - \gamma_2 - \frac{\sigma \beta_h \mu_h I_v^*}{B}} \left[\frac{1}{\frac{\sigma \beta_h \mu_h I_v^*}{B} + \gamma_2 - \gamma_1} \left(e^{-(\gamma_2 + \frac{\sigma \beta_h \mu_h I_v^*}{B})a} \right. \right. \right. \\ &\quad \left. \left. - e^{-\gamma_1 a} \right) - \frac{1}{k - \gamma_1} (e^{-ka} - e^{-\gamma_1 a}) \right] - \frac{1}{\frac{\beta_h \mu_h I_v^*}{B} - \frac{\sigma \beta_h \mu_h I_v^*}{B} - \gamma_2} \left[\frac{1}{\frac{\sigma \beta_h \mu_h I_v^*}{B} + \gamma_2 - \gamma_1} (-e^{-\gamma_1 a} \right. \right. \\ &\quad \left. \left. + e^{-(\frac{\sigma \beta_h \mu_h I_v^*}{B} + \gamma_2)a}) - \frac{1}{\frac{\beta_h \mu_h I_v^*}{B} - \gamma_1} (-e^{-\gamma_1 a} + e^{-\frac{\beta_h \mu_h I_v^*}{B} a}) \right] \right\}, \\ I_v^* &= \frac{\frac{\mu_h \beta_v}{B} \int_0^\infty P_\infty(a) (\sigma z_h^*(a) + i_h^*(a)) da}{\frac{\mu_h \beta_v}{B} \int_0^\infty P_\infty(a) (\sigma z_h^*(a) + I_h^*(a)) da + \mu_v} \in (0, 1), \\ \mathcal{R}_0 &= \frac{k \beta_v \beta_h \mu_h}{B \mu_v (\mu_h + k)} \frac{\rho (\mu_h + r_2) + \sigma (1 - \rho) (\mu_h + r_1)}{(\mu_h + r_1) (\mu_h + r_2)}. \end{aligned}$$

Substituting $z_h^*(a)$ and $i_h^*(a)$ into the equation for I_v^* , we obtain

$$H(I_v^*) \triangleq h_2 I_v^{*2} + h_1 I_v^* + h_0 = 0,$$

where,

$$\begin{aligned} h_2 &= \sigma k \beta_v \beta_h^2 \mu_h^3 + \sigma \mu_v \beta_h^2 \mu_h^2 (\mu_h + k) (\mu_h + \gamma_1) > 0, \\ h_1 &= B \mu_v \mu_h (\mu_h + k) (\mu_h + \gamma_1) (\sigma \beta_h \mu_h + (\mu_h + \gamma_2) \beta_h) + k B \beta_v \beta_h \mu_h^2 (\sigma (1 - \rho) (\mu_h + \gamma_1) \\ &\quad + \rho (\mu_h + \gamma_2)) - k \sigma \beta_v \beta_h^2 \mu_h^3, \\ h_0 &= B^2 \mu_v \mu_h (\mu_h + k) (\mu_h + \gamma_1) (\mu_h + \gamma_2) (1 - \mathcal{R}_0) > 0 \quad \text{when } \mathcal{R}_0 < 1. \end{aligned}$$

Obviously, the equation $H(I_v^*) = 0$ has two real roots $I_v^* \in (0, 1)$ if and only if the following conditions are satisfied:

$$\Delta = h_1^2 - 4h_0h_2 > 0, \quad 0 < -\frac{h_1}{2h_2} < 1.$$

Hence, if $\Delta > 0$ and $-2h_2 < h_1 < 0$, there are exactly two positive steady-state solutions when $\mathcal{R}_0 < 1$, and a backward bifurcation would occur.

6. Disease persistence

In this section, we study the system dynamics related to the disease persistence. To facilitate the analysis, we refocus our attention on system (2.1), and we expect the same persistence results to hold for system (2.2). As mentioned before, we assume that $P(a, t) = P_\infty(a)$ for sufficiently large t . Thus, $N_h(t) = N_h^*$. Since the equation for $R_h(a, t)$ is decoupled from the others, we only need to consider the reduced system:

$$\begin{aligned} \frac{\partial S_h(a, t)}{\partial a} + \frac{\partial S_h(a, t)}{\partial t} &= -\beta_h(a)I_v(t)\frac{S_h(a, t)}{N_h^*} - \mu_h S_h(a, t), \\ \frac{\partial E_h(a, t)}{\partial a} + \frac{\partial E_h(a, t)}{\partial t} &= \beta_h(a)I_v(t)\frac{S_h(a, t)}{N_h^*} - (\mu_h + k(a))E_h(a, t), \\ \frac{\partial I_h(a, t)}{\partial a} + \frac{\partial I_h(a, t)}{\partial t} &= \rho k(a)E_h(a, t) + \sigma\beta_h(a)I_v(t)\frac{A_h(a, t)}{N_h^*} - (\mu_h + \gamma_1(a))I_h(a, t), \\ \frac{\partial A_h(a, t)}{\partial a} + \frac{\partial A_h(a, t)}{\partial t} &= (1 - \rho)k(a)E_h(a, t) - \sigma\beta_h(a)I_v(t)\frac{A_h(a, t)}{N_h^*} - (\mu_h + \gamma_2(a))A_h(a, t), \\ \frac{dI_v(t)}{dt} &= (1 - I_v(t))\frac{1}{N_h^*} \int_0^\infty \beta_v(a)(I_h(a, t) + \sigma A_h(a, t))da - \mu_v I_v(t), \end{aligned} \quad (6.1)$$

equipped with the same boundary and initial conditions as (2.1). Let

$$\Omega_1 = \{(S_h, E_h, I_h, A_h, I_v)^T \in Y_+ : 0 \leq S_h + E_h + I_h + A_h \leq P_\infty, I_v \leq \frac{(1+\sigma)\bar{\beta}_v}{\mu_v + (1+\sigma)\bar{\beta}_v}\}.$$

With the same method used in Proposition 2.2, we can prove that system (6.1) has a unique mild solution in Ω_1 for any nonnegative initial condition. Furthermore, system (6.1) has a positively invariant set Ω_1 and generates a semiflow $\Phi(t) : \Omega_1 \rightarrow \Omega_1$,

$$\Phi[v_0](t) = (S_h(\cdot, t), E_h(\cdot, t), I_h(\cdot, t), A_h(\cdot, t), I_v(t)), \quad v_0 \in \Omega_1.$$

We call the disease uniformly weakly (respectively, strongly) endemic if there exists some $\epsilon > 0$ independent of the initial conditions such that

$$\limsup_{t \rightarrow \infty} I_v(t) > \epsilon \quad (\text{respectively, } \liminf_{t \rightarrow \infty} I_v(t) > \epsilon),$$

whenever $I_v(0) > 0$ for all solutions of system (6.1). We first prove the following proposition.

Proposition 6.1. *If $\mathcal{R}_0 > 1$, the disease is uniformly weakly endemic for system (6.1).*

Proof. We argue by contradiction. Assume $\limsup_{t \rightarrow \infty} I_v(t) \leq \epsilon$ for any $\epsilon > 0$. Then there exists a constant $T > 0$ such that for all $t > T$ we have $I_v(t) \leq \epsilon$. By shifting the dynamical system, we can assume that $I_v(t) \leq \epsilon$ for $t \geq 0$. Let

$$F_i(\theta, \xi, a) = k(\xi)\beta_h(\theta)e^{-\int_\theta^\xi [k(\phi) + \mu_h]d\phi}e^{-\int_\xi^a [\gamma_i(\phi) + \mu_h]d\phi}, \quad i = 1, 2.$$

Using the method of the characteristic lines for system (6.1), we have that for $t \geq a$,

$$\begin{aligned} A_h(a, t) &= \frac{B(1-\rho)}{N_h^*} \int_0^a \int_0^\xi k(\xi) \beta_h(\theta) I_v(t-a+\theta) e^{-\int_0^\theta [\frac{\beta_h(\phi) I_v(t-a+\phi)}{N_h^*} + \mu_h] d\phi} e^{-\int_\theta^\xi [k(\phi) + \mu_h] d\phi} \\ &\quad \times e^{-\int_\xi^a [\frac{\sigma \beta_h(\phi) I_v(t-a+\phi)}{N_h^*} + \gamma_2(\phi)] d\phi} d\theta d\xi, \\ I_h(a, t) &= \int_0^a \left(B\rho k(\varepsilon) \int_0^\varepsilon \frac{\beta_h(\theta) I_v(t-a+\theta)}{N_h^*} e^{-\int_0^\theta [\frac{\beta_h(\phi) I_v(t-a+\phi)}{N_h^*} + \mu_h] d\phi} e^{-\int_\theta^\varepsilon [k(\phi) + \mu_h] d\phi} d\theta \right. \\ &\quad + \frac{\sigma \beta_h(\varepsilon) I_v(t-a+\varepsilon)}{N_h^*} B(1-\rho) \int_0^\varepsilon \int_0^\xi k(\xi) \frac{\beta_h(\theta) I_v(t-a+\theta)}{N_h^*} e^{-\int_0^\theta [\frac{\beta_h(\phi) I_v(t-a+\phi)}{N_h^*} + \mu_h] d\phi} \\ &\quad \times e^{-\int_\theta^\xi [k(\phi) + \mu_h] d\phi} e^{-\int_\xi^\varepsilon [\frac{\sigma \beta_h(\phi) I_v(t-a+\phi)}{N_h^*} + \gamma_2(\phi) + \mu_h] d\phi} d\theta d\xi \Big) e^{-\int_\varepsilon^a [\gamma_1(\phi) + \mu_h] d\phi} d\varepsilon. \end{aligned}$$

From the last equation in system (6.1) and the expressions above for $I_h(a, t)$ and $A_h(a, t)$, we have

$$\begin{aligned} \frac{dI_v}{dt} &= \frac{1-I_v(t)}{N_h^*} \int_0^\infty \beta_v(a) (I_h(a, t) + \sigma A_h(a, t)) da - \mu_v I_v(t) \\ &\geq \frac{1-\epsilon}{N_h^*} \int_0^t \beta_v(a) (I_h(a, t) + \sigma A_h(a, t)) da - \mu_v I_v(t) \\ &\geq \frac{1-\epsilon}{N_h^*} \int_0^t \beta_v(a) \left(\frac{B\rho}{N_h^*} \int_0^a \int_0^\xi k(\xi) \beta_h(\theta) I_v(t-a+\theta) e^{-\int_0^\theta [\frac{\beta_h(\phi) \epsilon}{N_h^*} + \mu_h] d\phi} e^{-\int_\theta^\xi [k(\phi) + \mu_h] d\phi} \right. \\ &\quad \times e^{-\int_\xi^a [\gamma_1(\phi) + \mu_h] d\phi} d\theta d\xi + \frac{\sigma B(1-\rho)}{N_h^*} \int_0^a \int_0^\xi k(\xi) \beta_h(\theta) I_v(t-a+\theta) e^{-\int_0^\theta [\frac{\beta_h(\phi) \epsilon}{N_h^*} + \mu_h] d\phi} \\ &\quad \times e^{-\int_\theta^\xi [k(\phi) + \mu_h] d\phi} e^{-\int_\xi^a [\gamma_2(\phi) + \mu_h] d\phi} e^{-\int_\xi^a \frac{\sigma \beta_h(\phi) \epsilon}{N_h^*} d\phi} d\theta d\xi \Big) da - \mu_v I_v(t). \end{aligned}$$

Setting

$$H_1(\theta) = e^{-\int_0^\theta [\frac{\beta_h(\phi) \epsilon}{N_h^*} + \mu_h] d\phi}, \quad H_2(\xi, a) = e^{-\int_\xi^a \frac{\sigma \beta_h(\phi) \epsilon}{N_h^*} d\phi},$$

we obtain

$$\begin{aligned} \frac{dI_v}{dt} &\geq \frac{1-\epsilon}{N_h^*} \int_0^t \beta_v(a) \left(\frac{B\rho}{N_h^*} \int_0^a \int_0^\xi I_v(t-a+\theta) F_1(\theta, \xi, a) H_1(\theta) d\theta d\xi \right. \\ &\quad \left. + \frac{\sigma B(1-\rho)}{N_h^*} \int_0^a \int_0^\xi I_v(t-a+\theta) F_2(\theta, \xi, a) H_1(\theta) H_2(\xi, a) d\theta d\xi \right) da - \mu_v I_v. \end{aligned}$$

Next, we take the Laplace transform on both sides of the inequalities above and denote by $\hat{I}_v$ the Laplace transform of $I_v(t)$. Note that

$$\begin{aligned}
& \int_0^\infty e^{-\lambda t} \int_0^t \beta_v(a) \frac{B\rho}{N_h^*} \int_0^a \int_0^\xi I_v(t-a+\theta) F_1(\theta, \xi, a) H_1(\theta) d\theta d\xi da dt \\
&= \int_0^\infty da \int_a^\infty e^{-\lambda t} \beta_v(a) \frac{B\rho}{N_h^*} \int_0^a \int_0^\xi I_v(t-a+\theta) F_1(\theta, \xi, a) H_1(\theta) d\theta d\xi dt \\
&= \int_0^\infty da \int_0^\infty e^{-\lambda(t+a)} \beta_v(a) \frac{B\rho}{N_h^*} \int_0^a \int_0^\xi I_v(t+\theta) F_1(\theta, \xi, a) H_1(\theta) d\theta d\xi dt \\
&= \hat{I}_v(\lambda) \int_0^\infty e^{-\lambda a} \beta_v(a) \frac{B\rho}{N_h^*} \int_0^a \int_0^\xi F_1(\theta, \xi, a) H_1(\theta) d\theta d\xi da.
\end{aligned}$$

Similarly, we can obtain

$$\begin{aligned}
& \int_0^\infty e^{-\lambda t} \int_0^t \beta_v(a) \frac{\sigma B(1-\rho)}{N_h^*} \int_0^a \int_0^\xi I_v(t-a+\theta) F_2(\theta, \xi, a) H_1(\theta) H_2(\xi, a) d\theta d\xi da dt \\
&= \hat{I}_v(\lambda) \int_0^\infty e^{-\lambda a} \beta_v(a) \frac{\sigma B(1-\rho)}{N_h^*} \int_0^a \int_0^\xi F_2(\theta, \xi, a) H_1(\theta) H_2(\xi, a) d\theta d\xi da.
\end{aligned}$$

Thus, we have

$$\begin{aligned}
\lambda \hat{I}_v(\lambda) - I_v(0) &\geq \hat{I}_v(\lambda) \frac{1-\epsilon}{N_h^*} \int_0^\infty e^{-\lambda a} \beta_v(a) \frac{B\rho}{N_h^*} \int_0^a \int_0^\xi F_1(\theta, \xi, a) H_1(\theta) d\theta d\xi da - \mu_v \hat{I}_v(\lambda) \\
&\quad + \hat{I}_v(\lambda) \frac{1-\epsilon}{N_h^*} \int_0^\infty e^{-\lambda a} \beta_v(a) \frac{\sigma B(1-\rho)}{N_h^*} \int_0^a \int_0^\xi F_2(\theta, \xi, a) H_1(\theta) H_2(\xi, a) d\theta d\xi da \\
&= \hat{I}_v(\lambda) \frac{1-\epsilon}{N_h^*} \int_0^\infty e^{-\lambda a} \beta_v(a) \left(\frac{B\rho}{N_h^*} \int_0^a \int_0^\xi F_1(\theta, \xi, a) H_1(\theta) d\theta d\xi \right. \\
&\quad \left. + \frac{\sigma B(1-\rho)}{N_h^*} \int_0^a \int_0^\xi F_2(\theta, \xi, a) H_1(\theta) H_2(\xi, a) d\theta d\xi \right) da - \mu_v \hat{I}_v(\lambda) \\
&= \hat{I}_v(\lambda) \frac{1-\epsilon}{N_h^*} \int_0^\infty e^{-\lambda a} \beta_v(a) P_\infty(a) \left(\frac{\rho}{N_h^*} \int_0^a \int_0^\xi k(\xi) \beta_h(\theta) e^{-\int_\theta^\xi k(\phi) d\phi} e^{-\int_\xi^a \gamma_1(\phi) d\phi} \right. \\
&\quad \times e^{-\int_0^\theta \frac{\beta_h(\phi)\epsilon}{N_h^*} d\phi} d\theta d\xi + \frac{\sigma(1-\rho)}{N_h^*} \int_0^a \int_0^\xi k(\xi) \beta_h(\theta) e^{-\int_\theta^\xi k(\phi) d\phi} e^{-\int_\xi^a \gamma_2(\phi) d\phi} e^{-\int_0^\theta \frac{\beta_h(\phi)\epsilon}{N_h^*} d\phi} \\
&\quad \left. \times e^{-\int_\xi^a \frac{\sigma \beta_h(\phi)\epsilon}{N_h^*} d\phi} d\theta d\xi \right) da - \mu_v \hat{I}_v(\lambda).
\end{aligned}$$

We can rewrite the above expression as

$$\begin{aligned} \hat{I}_v(\lambda) &\geq \hat{I}_v(\lambda) \frac{1-\epsilon}{N_h^*(\lambda+\mu_v)} \int_0^\infty e^{-\lambda a} \beta_v(a) P_\infty(a) \left(\frac{\rho}{N_h^*} \int_0^a \int_0^\xi k(\xi) \beta_h(\theta) e^{-\int_\theta^\xi k(\phi) d\phi} e^{-\int_\xi^a \gamma_1(\phi) d\phi} \right. \\ &\quad \times e^{-\int_0^\theta \frac{\beta_h(\phi)\epsilon}{N_h^*} d\phi} d\theta d\xi + \frac{\sigma(1-\rho)}{N_h^*} \int_0^a \int_0^\xi k(\xi) \beta_h(\theta) e^{-\int_\theta^\xi k(\phi) d\phi} e^{-\int_\xi^a \gamma_2(\phi) d\phi} e^{-\int_0^\theta \frac{\beta_h(\phi)\epsilon}{N_h^*} d\phi} \\ &\quad \left. \times e^{-\int_\xi^a \frac{\sigma\beta_h(\phi)\epsilon}{N_h^*} d\phi} d\theta d\xi \right) da + \frac{I_v(0)}{\lambda+\mu_v}. \end{aligned}$$

The inequality above should hold for the given $\epsilon \approx 0$ and any $\lambda > 0$. However, this is impossible because the coefficient of $\hat{I}_v(\lambda)$ on the right-hand side is approximately $\mathcal{R}_0 > 1$. Besides, there is another positive term on the right-hand side of the inequality. This contradiction shows that the disease is uniformly weakly endemic. $\square$

Proposition 6.2. *If $\mathcal{R}_0 > 1$, then there exists a global compact attractor for the solution semiflow Φ of system (6.1).*

Proof. To establish this result, we will use Lemma 3.2.3 and Theorem 3.4.6 in [14]. There methods and techniques have been recently employed in [6]. To this end, we split the solution semiflow Φ into two components. For any initial condition x_0 , we let $\Phi(t, x_0) = \hat{\Phi}(t, x_0) + \tilde{\Phi}(t, x_0)$ such that the following assertions hold:

- (1) $\hat{\Phi}(t, x_0) \rightarrow 0$ as $t \rightarrow \infty$ for every x_0 ;
- (2) For a fixed t and any bounded set M_0 , the set $\{\tilde{\Phi}(t, x_0) : x_0 \in M_0\}$ is precompact.

Note that the two components of the semiflow of the limiting system are, respectively, defined as follows:

$$\begin{aligned} \hat{\Phi}(t, S_{h0}(\cdot), E_{h0}(\cdot), I_{h0}(\cdot), A_{h0}(\cdot), I_{v0}) &= (\hat{S}_h(a, t), \hat{E}_h(a, t), \hat{I}_h(a, t), \hat{A}_h(a, t), 0), \\ \tilde{\Phi}(t, S_{h0}(\cdot), E_{h0}(\cdot), I_{h0}(\cdot), A_{h0}(\cdot), I_{v0}) &= (\tilde{S}_h(a, t), \tilde{E}_h(a, t), \tilde{I}_h(a, t), \tilde{A}_h(a, t), I_v(t)), \end{aligned}$$

where

$$\hat{S}_h(a, t) = \begin{cases} 0, & a \leq t, \\ S_{h0}(a-t) e^{-\int_0^t [\frac{\beta_h(a-t+\phi)I_v(\phi)}{N_h^*} + \mu_h] d\phi}, & t < a, \end{cases} \quad (6.2)$$

and

$$\tilde{S}_h(a, t) = \begin{cases} B e^{-\int_0^a [\frac{\beta_h(\phi)I_v(t-a+\phi)}{N_h^*} + \mu_h] d\phi}, & a \leq t, \\ 0, & t < a. \end{cases} \quad (6.3)$$

Integrating $\hat{S}_h(a, t)$ with respect to a in the interval $[0, \infty)$ yields

$$\begin{aligned} \int_0^\infty \hat{S}_h(a, t) da &= \int_t^\infty \hat{S}_h(a, t) da = \int_t^\infty S_{h0}(a-t) e^{-\int_0^t [\frac{\beta_h(a-t+\phi)I_v(\phi)}{N_h^*} + \mu_h] d\phi} da \\ &\leq \int_t^\infty S_{h0}(a-t) e^{-\mu_h t} da = e^{-\mu_h t} \int_0^\infty S_{h0}(a) da \rightarrow 0 \quad (t \rightarrow \infty). \end{aligned} \quad (6.4)$$

Using a similar argument, we can obtain

$$\begin{aligned}
 \int_0^\infty \hat{E}_h(a, t) da &\leq e^{-\mu_h t} \left(\int_0^\infty E_{h0}(a) da + \frac{\bar{\beta}_h t}{N_h^*} \int_0^\infty S_{h0}(a) da \right) \rightarrow 0 \quad (t \rightarrow \infty), \\
 \int_0^\infty \hat{A}_h(a, t) da &\leq e^{-\mu_h t} \left(\int_0^\infty A_{h0}(a) da + \bar{k}(1-\rho)t \int_0^\infty E_{h0}(a) da \right. \\
 &\quad \left. + \frac{\bar{k}(1-\rho)\bar{\beta}_h t^2}{2N_h^*} \int_0^\infty S_{h0}(a) da \right) \rightarrow 0 \quad (t \rightarrow \infty), \\
 \int_0^\infty \hat{I}_h(a, t) da &\leq e^{-\mu_h t} \left(\int_0^\infty I_{h0}(a) da + \rho \bar{k} t \int_0^\infty E_{h0}(a) da + \frac{\rho \bar{k} \bar{\beta}_h t^2}{2N_h^*} \int_0^\infty S_{h0}(a) da \right. \\
 &\quad \left. + \frac{\sigma \bar{\beta}_h t}{N_h^*} \int_0^\infty A_{h0}(a) da + \frac{\sigma \bar{\beta}_h \bar{k}(1-\rho)t^2}{2N_h^*} \int_0^\infty E_{h0}(a) da \right) \rightarrow 0 \quad (t \rightarrow \infty).
 \end{aligned} \tag{6.5}$$

Now we show that the first assertion above is true. In fact, it is easy to know that $\hat{S}_h(a, t)$, $\hat{E}_h(a, t)$, $\hat{A}_h(a, t)$, $\hat{I}_h(a, t)$ are nonnegative. It follows from (6.4) and (6.5) that

$$\begin{aligned}
 \|\Phi\|_X &= \|\hat{S}_h(\cdot, t)\|_{L^1} + \|\hat{E}_h(\cdot, t)\|_{L^1} + \|\hat{A}_h(\cdot, t)\|_{L^1} + \|\hat{I}_h(\cdot, t)\|_{L^1} \\
 &= \int_0^\infty \hat{S}_h(a, t) da + \int_0^\infty \hat{E}_h(a, t) da + \int_0^\infty \hat{A}_h(a, t) da + \int_0^\infty \hat{I}_h(a, t) da \\
 &\leq e^{-\mu_h t} \left(\int_0^\infty S_{h0}(a) da + \int_0^\infty E_{h0}(a) da + \frac{\bar{\beta}_h t}{N_h^*} \int_0^\infty S_{h0}(a) da + \int_0^\infty A_{h0}(a) da \right. \\
 &\quad + \bar{k}(1-\rho)t \int_0^\infty E_{h0}(a) da + \frac{\bar{k}(1-\rho)\bar{\beta}_h t^2}{2N_h^*} \int_0^\infty S_{h0}(a) da + \int_0^\infty I_{h0}(a) da \\
 &\quad + \rho \bar{k} t \int_0^\infty E_{h0}(a) da + \frac{\rho \bar{k} \bar{\beta}_h t^2}{2N_h^*} \int_0^\infty S_{h0}(a) da + \frac{\sigma \bar{\beta}_h \bar{k}(1-\rho)t^2}{2N_h^*} \int_0^\infty E_{h0}(a) da \\
 &\quad \left. + \frac{\sigma \bar{\beta}_h t}{N_h^*} \int_0^\infty A_{h0}(a) da \right) \rightarrow 0 \text{ as } t \rightarrow \infty.
 \end{aligned}$$

Next, we need to justify the second assertion. By the previously defined set $\Omega \subset X_0$, it is ready to obtain that the family $\{\tilde{\Phi}(t, x_0) : x_0 \in X_0, t \text{ fixed}\} \in X_0$ is bounded. To prove the compactness, we need to show the remaining conditions to hold in the Fréchet-Kolmogorov theorem [39]. The first step is to show that $\tilde{S}_h(a, t)$ satisfies the third condition of the Fréchet-Kolmogorov theorem. This is obvious since

$$\tilde{S}_h(a, t) = 0 \quad \text{for } a > t. \tag{6.6}$$

Now we only need to show the following:

$$\lim_{h \rightarrow 0^+} \int_0^\infty |\tilde{S}_h(a+h, t) - \tilde{S}_h(a, t)| da = 0, \quad \text{uniformly in some bounded set } \Omega_0 \subset X_0.$$

From the equation for $I_v(t)$ in the limiting system,

$$\begin{aligned} \frac{dI_v(t)}{dt} &\leq \frac{1}{N_h^*} \int_0^\infty \beta_v(a)(I_h(a, t) + \sigma A_h(a, t)) da + \mu_v \\ &\leq \frac{\bar{\beta}_v}{N_h^*} \int_0^\infty (I_h(a, t) + \sigma A_h(a, t)) da + \mu_v \\ &\leq \frac{\bar{\beta}_v}{N_h^*} (N_h^* + \sigma N_h^*) + \mu_v = \bar{\beta}_v(1 + \sigma) + \mu_v \triangleq c_0. \end{aligned}$$

To estimate the bound of $\frac{\partial \tilde{S}_h(a, t)}{\partial a}$, we differentiate (6.3) with respect to a and obtain

$$\left| \frac{\tilde{S}_h(a, t)}{\partial a} \right| = \begin{cases} 0, & t < a, \\ Be^{-\mu_h a} \left(\frac{\bar{\beta}_h}{N_h^*} + \mu_h + \frac{\bar{\beta}_h c_0}{N_h^*} a \right), & a \leq t. \end{cases} \quad (6.7)$$

Hence,

$$\begin{aligned} \left\| \frac{\tilde{S}_h(a, t)}{\partial a} \right\|_{L^1} &= \int_0^t \left| \frac{\tilde{S}_h(a, t)}{\partial a} \right| da + \int_t^\infty \left| \frac{\tilde{S}_h(a, t)}{\partial a} \right| da = \int_0^t Be^{-\mu_h a} \left(\frac{\bar{\beta}_h}{N_h^*} + \mu_h + \frac{\bar{\beta}_h c_0}{N_h^*} a \right) da \\ &\leq B \int_0^\infty e^{-\mu_h a} \left(\frac{\bar{\beta}_h}{N_h^*} + \mu_h + \frac{\bar{\beta}_h c_0}{N_h^*} a \right) da = B \left(\frac{\bar{\beta}_h}{\mu_h N_h^*} + 1 + \frac{\bar{\beta}_h c_0}{\mu_h^2 N_h^*} \right) \triangleq c_1. \end{aligned}$$

Also notice that

$$\int_0^\infty |\tilde{S}_h(a+h, t) - \tilde{S}_h(a, t)| da \leq \left\| \frac{\tilde{S}_h(a, t)}{\partial a} \right\|_{L^1} |h| \leq c_1 |h|. \quad (6.8)$$

Obviously, this integral can be arbitrarily small uniformly in the family of functions. Similar results can be established for $\tilde{E}_h(a, t)$, $\tilde{A}_h(a, t)$ and $\tilde{I}_h(a, t)$:

$$\tilde{E}_h(a, t) = \tilde{A}_h(a, t) = \tilde{I}_h(a, t) = 0, \quad \text{for } a > t, \quad (6.9)$$

$$\int_0^\infty |\tilde{E}_h(a+h, t) - \tilde{E}_h(a, t)| da \leq \left\| \frac{\tilde{E}_h(a, t)}{\partial a} \right\|_{L^1} |h| \leq c_2 |h|, \quad (6.10)$$

$$\int_0^\infty |\tilde{A}_h(a+h, t) - \tilde{A}_h(a, t)| da \leq \left\| \frac{\tilde{A}_h(a, t)}{\partial a} \right\|_{L^1} |h| \leq c_3 |h|, \quad (6.11)$$

$$\int_0^\infty |\tilde{I}_h(a+h, t) - \tilde{I}_h(a, t)| da \leq \left\| \frac{\tilde{I}_h(a, t)}{\partial a} \right\|_{L^1} |h| \leq c_4 |h|, \quad (6.12)$$

where each c_i ($i = 2, 3, 4$) is a constant which depends on the bounds of the model parameters. The equations (6.6) and (6.9) show that the third condition in the Fréchet-Kolmogorov theorem holds. Meanwhile, the inequalities (6.8), (6.10), (6.11) and (6.12) imply the second condition holds. In addition, the first condition regarding the boundedness has already been verified. Hence, all the three conditions in the Fréchet-Kolmogorov theorem are satisfied. The proof is thus completed. $\square$

Based on Propositions 6.1 and 6.2, and Theorem 2.6 in [34] (with similar methods recently employed in [23]), we immediately obtain the following result:

Theorem 6.3. *If $\mathcal{R}_0 > 1$, the disease is uniformly strongly endemic for system (6.1).*

7. Conclusions

We have presented a novel age-structured model to investigate the transmission dynamics of dengue. Our model links the interaction between the human hosts of different ages and mosquitoes, includes both the symptomatic and asymptomatic classes, and incorporates the effects of primary and secondary infections. We have placed a special emphasis on the asymptomatic individuals and their interaction with mosquitoes and other human hosts. Our results highlight the roles of the asymptomatic hosts and secondary infection in shaping dengue transmission under a heterogeneous setting that couples the host and vector populations and that involves different age groups.

A detailed mathematical analysis has been conducted for this model using the basic reproduction number $\mathcal{R}_0$ derived in Section 3. It is proved that the disease-free steady state is locally asymptotically stable if $\mathcal{R}_0 < 1$ and globally asymptotically stable if $\mathcal{R}_0 < 1 - \varsigma$ for some $\varsigma > 0$. It is also shown that when $\mathcal{R}_0 > 1$, there exists at least one endemic steady state and the disease is both weakly and strongly uniformly persistent. Numerical simulation results confirm these analytical predictions. Additionally, sufficient conditions are established for the existence of multiple endemic steady states when $\mathcal{R}_0 < 1$ which could lead to a backward bifurcation.

The threshold dynamics results indicate that to eradicate the disease, it may not be sufficient to simply reduce the basic reproduction number $\mathcal{R}_0$ below unity. Instead, control measures should be made to further reduce $\mathcal{R}_0$ so that it is outside the interval for the possible backward bifurcation and falls within the region of global asymptotic stability for the disease-free steady state. The expression for $\mathcal{R}_0$ includes contributions from both the human and vector populations, underscoring the importance of controlling both the infected hosts and mosquitoes in the management of dengue. Meanwhile, the equation of $\mathcal{R}_0$ represents the two transmission routes associated with the symptomatic and asymptomatic infectious hosts. Therefore, dengue control strategies should target both symptomatic and asymptomatic individuals to effectively reduce the value of $\mathcal{R}_0$. In particular, reducing the reinfection intensity factor σ would directly decrease the contribution of the asymptomatic individuals to $\mathcal{R}_0$. If new diagnostic testing methods, such as those for COVID-19, can be developed and widely implemented to rapidly identify people infected with dengue, especially those asymptomatic individuals, that would substantially reduce the transmission risk of dengue. Moreover, our model and results also build a foundation for a study of optimal control strategies for dengue based on different age groups.

The transmission dynamics of dengue are complex, and there are many factors that are potentially important and that have not been considered in the present work. These include, but are not limited to, delay from actual infection to host awareness of the infection, effect of ADE on the transmission rates, heterogeneity in the spatial spread of the disease, and impact of dengue vaccination. We plan to study these topics in our future research.

Acknowledgments

Authors are very grateful to the anonymous referees for their careful reading, helpful comments and suggestions, which have helped us to improve the presentation of this work significantly.

References

- [1] M. Aguiar, S. Ballesteros, B.W. Kooi, N. Stollenwerk, The role of seasonality and import in a minimalistic multi-strain dengue model capturing differences between primary and secondary infections: complex dynamics and its implications for data analysis, *J. Theor. Biol.* 289 (2011) 181–196.
- [2] M. Aguiar, V. Anam, K.B. Blyuss, C.D.S. Estadilla, B.V. Guerrero, D. Knopoff, et al., Mathematical models for dengue fever epidemiology: a 10-year systematic review, *Phys. Life Rev.* 40 (2022) 65–92.
- [3] S. Bhatt, P.W. Gething, O.J. Brady, et al., The global distribution and burden of dengue, *Nature* 496 (7446) (2013) 504–507.
- [4] S.N. Busenberg, M. Iannelli, H.R. Thieme, Global behavior of an age-structured epidemic model, *SIAM J. Math. Anal.* 22 (4) (1991) 1065–1080.
- [5] L.M. Cai, X.Z. Li, X.Y. Song, et al., Permanence and stability of an age-structured prey-predator system with delays, *Discrete Dyn. Nat. Soc.* 2007 (2007) 149–174.
- [6] L.M. Cai, Z.Q. Li, C.Y. Yang, J. Wang, Global analysis of an environmental disease transmission model linking within-host and between-host dynamics, *Appl. Math. Model.* 86 (2020) 404–423.
- [7] L.M. Cai, N. Tuncer, M. Martcheva, How does within-host dynamics affect population-level dynamics? Insights from an immuno-epidemiological model of malaria, *Math. Methods Appl. Sci.* 40 (18) (2017) 6424–6450.
- [8] G.F. Chong, L.M. Cai, Analysis for an age-structured SVIR epidemic model with vaccination, in: *Proceedings of the 31st Chinese Control Conference, IEEE, 2012*, pp. 1296–1300.
- [9] G. Cruz-Pacheco, L. Esteva, C. Vargas, Vaccination strategies for SIR vector-transmitted diseases, *Bull. Math. Biol.* 76 (8) (2014) 2073–2090.
- [10] C. Ding, Z. Wang, Q. Zhang, Age-structure model for oncolytic virotherapy, *Int. J. Biomath.* 15 (01) (2022) 2150091.
- [11] L. Esteva, C. Vargas, Coexistence of different serotypes of dengue virus, *J. Math. Biol.* 46 (2003) 31–47.
- [12] J.Z. Farkas, S.A. Gourley, R. Liu, Dengue transmission dynamics in age-structured human populations in the presence of Wolbachia, *arXiv:2111.04399*, available at <https://doi.org/10.48550/arXiv.2111.04399>, November 2021.
- [13] N. Ferguson, R. Anderson, S. Gupta, The effect of antibody-dependent enhancement on the transmission dynamics and persistence of multiple-strain pathogens, *Proc. Natl. Acad. Sci.* 96 (2) (1999) 790–794.
- [14] J.K. Hale, *Asymptotic Behavior of Dissipative Systems*, AMS, Providence, 1988.
- [15] E. Harris, E. Videa, L. Pérez, et al., Clinical, epidemiologic, and virologic features of dengue in the 1998 epidemic in Nicaragua, *Am. J. Trop. Med. Hyg.* 63 (1–2) (2000) 5–11.
- [16] C.Q. He, Y. Zhao, Stability analysis of a dengue thermokinetic model with latent effects, *Adv. Appl. Math.* 10 (7) (2021) 2472–2485.
- [17] W.M. Hirsch, H. Hanisch, J.-P. Gabriel, Differential equation models of some parasitic infections: methods for the study of asymptotic behavior, *Commun. Pure Appl. Math.* 38 (6) (1985) 733–753.
- [18] R. Jan, M.A. Khan, J.F. Gómez-Aguilar, Asymptomatic carriers in transmission dynamics of dengue with control interventions, *Optim. Control Appl. Methods* 41 (2) (2020) 430–447.
- [19] R. Jan, Y. Xiao, Effect of partial immunity on transmission dynamics of dengue disease with optimal control, *Math. Methods Appl. Sci.* 42 (6) (2019) 1967–1983.
- [20] K.A. Kabir, J. Tanimoto, Cost-efficiency analysis of voluntary vaccination against n-serovar diseases using antibody-dependent enhancement: a game approach, *J. Theor. Biol.* 503 (2020) 110379.
- [21] X.Z. Li, G. Gupur, Global stability of an age-structured SIRS epidemic model with vaccination, *Discrete Contin. Dyn. Syst., Ser. B* 4 (3) (2004) 643–652.
- [22] S.B. Maier, E. Massad, M. Amaku, et al., The optimal age of vaccination against dengue with an age-dependent biting rate with application to Brazil, *Bull. Math. Biol.* 82 (1) (2020) 1–32.
- [23] M. Martcheva, H.R. Thieme, Progression age enhanced backward bifurcation in an epidemic model with super-infection, *J. Math. Biol.* 46 (5) (2003) 385–424.
- [24] A. Mishra, S. Gakkhar, Modeling of dengue with impact of asymptomatic infection and ADE factor, *Differ. Equ. Dyn. Syst.* 28 (3) (2020) 745–761.
- [25] K. Okuwa, H. Inaba, T. Kuniya, Mathematical analysis for an age-structured SIRS epidemic model, *Math. Biosci. Eng.* 16 (5) (2019) 6071–6102.
- [26] P. Pongsumpun, D. Samana, Mathematical model for asymptomatic and symptomatic infections of dengue disease, *WSEAS Trans. Biol. Biomed.* 3 (3) (2006) 49–54.
- [27] P. Pongsumpun, I.M. Tang, Transmission of dengue hemorrhagic fever in an age structured population, *Math. Comput. Model.* 37 (2003) 949–961.
- [28] S. Ranjit, N. Kissoon, Dengue hemorrhagic fever and shock syndromes, *Ped. Crit. Care Med.* 12 (1) (2011) 90–100.
- [29] P. Rashkov, B.W. Kooi, Complexity of host-vector dynamics in a two-strain dengue model, *J. Biol. Dyn.* 15 (1) (2021) 35–72.
- [30] F. Rocha, L. Mateus, U. Skwara, M. Aguiar, N. Stollenwerk, Understanding dengue fever dynamics: a study of seasonality in vector-borne disease models, *Int. J. Comput. Math.* 93 (8) (2016) 1405–1422.

- [31] F. Sanchez, J.G. Calvo, Dengue model with early-life stage of vectors and age-structure within host, *Rev. Mat. Teor. Apl.* 27 (1) (2020) 157–177.
- [32] F. Sanchez, J.G. Calvo, E. Segura, et al., A partial differential equation model with age-structure and nonlinear recidivism: conditions for a backward bifurcation and a general numerical implementation, *Comput. Math. Appl.* 78 (12) (2019) 3916–3930.
- [33] C.P. Simmons, J.J. Farrar, N. van Vinh Chau, B. Wills, Dengue, *N. Engl. J. Med.* 366 (15) (2012) 1423–1432.
- [34] H.R. Thieme, Uniform persistence and permanence for non-autonomous semiflows in population biology, *Math. Biosci.* 166 (2) (2000) 173–201.
- [35] X. Tian, W.D. Wang, Dynamical analysis of age-structured pertussis model with covert infection, *Math. Methods Appl. Sci.* 43 (4) (2020) 1631–1645.
- [36] World Health Organization, Dengue and severe dengue, available at <https://www.who.int/en/news-room/fact-sheets/detail/dengue-and-severe-dengue>.
- [37] Z. Xie, X. Liu, Global dynamics in an age-structured HIV model with humoral immunity, *Int. J. Biomath.* 14 (06) (2021) 2150047.
- [38] J.Y. Yang, C. Mondnak, J. Wang, Dynamical analysis and optimal control simulation for an age-structured cholera transmission model, *J. Franklin Inst.* 356 (15) (2019) 8438–8467.
- [39] K. Yosida, *Functional Analysis*, second edition, Springer-Verlag, 1968.